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On the equation $N_{K/k}(\Xi) = P(t)$

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Abstract

For varieties given by an equation $N_{K/k}(\Xi) = P(t)$, where $N_{K/k}$ is the norm form attached to a field extension K/k and $P(t)$ in $k[t]$ is a non-zero polynomial, three topics have been investigated:

(i) computation of the unramified Brauer group of such varieties over arbitrary fields;
(ii) rational points and Brauer–Manin obstruction over number fields (under Schinzel's hypothesis);
(iii) zero-cycles and Brauer–Manin obstruction over number fields.

In this paper, we produce new results in each of three directions. We obtain quite general results under the assumption that K/k is abelian (as opposed to cyclic in earlier investigation).

1. *Introduction*

A class of geometrically integral varieties defined over a number field k satisfies the Hasse principle if a variety in this class has a k-rational point as soon as it has rational points in all the completions k_v of the field k. For example, quadrics, Severi–Brauer varieties are known to satisfy this principle. However, counterexamples to the Hasse principle are also known even in the class of geometrically rational varieties. In 1970, Manin [**29**, **30**] showed that an obstruction based on the Brauer group of varieties, now referred to as the Brauer–Manin obstruction, can often explain failures of the Hasse principle. Further work (see [**5**] for a survey) has shown that for some classes of rational varieties the Brauer–Manin obstruction is the only obstruction to the Hasse principle.

For some classes of varieties for which the Hasse principle has been proved, weak approximation is also known: namely, given a variety X over a number field k, and given a k_v-rational point P_v of X for each v in a finite set of places of k, one may find a k-rational point on X as close as one wishes to each P_v (for the v-adic topology). However, for more general unirational varieties, counterexamples to weak approximation are known, and one may define a Brauer–Manin obstruction to weak approximation [**11**, **12**, **15**] and ask whether it is the only obstruction in the class of geometrically unirational varieties.

For general varieties, it seems quite unreasonable to hope for such statements. A more reasonable conjecture relates to zero-cycles of degree 1. A variety defined over a field k has a zero-cycle of degree 1 over k if and only if the degrees of the finite field extensions K/k over which it acquires a rational point are globally coprime. There also exist counterexamples to the Hasse principle for zero-cycles of degree 1. Similarly, one may define a Brauer–Manin obstruction and ask whether this is the only obstruction for arbitrary smooth projective varieties over a number field [**6**].

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The present paper focuses on varieties defined over the ground field k by an equation

$$N_{K/k}(\Xi) = P(t), \tag{1.1}$$

where K/k is a finite field extension, $N_{K/k}$ denotes the norm map, Ξ is a 'variable' in K and $P(t) \in k[t]$ is a non-constant polynomial.

To compute the Brauer–Manin obstruction, one must compute the Brauer group of a smooth projective model of the variety under study. That group is sometimes referred to as the unramified Brauer group of the variety.

In 2003, Colliot-Thélène, Harari and Skorobogatov [**8**] discussed the unramified Brauer group for these varieties. In fact, they defined a partial compactification of the smooth locus of these varieties. For this partial compactification, they gave a formula for its vertical Brauer group and the quotient of its Brauer group by the vertical Brauer group. They pointed out that sometimes the unramified Brauer group can be calculated by the formula. However, it is still open how to determine the unramified Brauer group for more general cases.

Let X be the partial compactification of the smooth locus of the affine variety defined by (1.1) (see [**8**]), X^c its smooth compactification. In Section 2, building upon the formula for $\mathrm{Br}(X)$ in [**8**], we compute $\mathrm{Br}(X^c)$ in several new cases.

(i) In Section 3.1, for $P(t)$ irreducible and K/k 'general', we show $\mathrm{Br}(X^c) = \mathrm{Br}_0(X^c)$ (Theorem 3.2).

(ii) In Section 3.2, for $P(t)$ irreducible and K/k abelian, we show that the quotient $\mathrm{Br}(X)/\mathrm{Br}(X^c)$ is 2-torsion, and we show that the quotient is trivial in many cases (Theorem 3.6).

(iii) In Section 3.3, motivated by the question in [**8**], we give a formula for $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$ when K/k is Galois and $P(t)$ has all roots in k, not necessarily distinct.

In a series of earlier papers, rational points and zero-cycles of degree 1 have been studied on smooth projective models of varieties defined by an equation $N_{K/k}(\Xi) = P(t)$, and more generally on varieties fibred over the projective line whose general fibre is birationally a principal homogeneous space under a torus.

Under Schinzel's hypothesis (H), the question was first studied by Colliot-Thélène and Sansuc [**13**]. Further work under (H) is due to Colliot-Thélène, Serre, Skorobogatov and Swinnerton-Dyer [**17**, **18**, **36**, **40**].

As explained in [**17**, **18**], a device due to Salberger [**33**, **34**] enables one to transform some of the conditional proofs for rational points obtained under (H) into unconditional proofs for zero-cycles of degree 1.

In [**17**], restrictions are made on the fibres: on the one hand, one assumes that the Hasse principle and weak approximation hold on all smooth fibres, on the other hand, one requires some abelianity condition for the splitting fields associated to the components of the singular fibres.

In this paper, we prove results of the above type in the cases not covered by [**17**]: *either the Hasse principle need not hold on the fibres, or the abelian splitting condition is not fulfilled.*

We consider three types of varieties over the ground field k:

(1) varieties defined by an equation $N_{K/k}(\Xi) = P(t)$, where K/k is an abelian extension and $Ш^2_\omega(\hat{T})_P = Ш^2_\omega(\hat{T})$ (see Section 1 for definition);
(2) varieties defined by an equation $(x_1^2 - ax_2^2)(y_1^2 - by_2^2)(z_1^2 - abz_2^2) = P(t)$, where $a, b \in k^*$;
(3) varieties defined by an equation $N_{K/k}(\Xi) = P(t)$, where K/k has prime degree (non-Galois) and $P(t)$ satisfies condition (4.10).

In Section 3, we prove the following theorem.

THEOREM 1.1. *Let V be the smooth locus of one of the above three varieties. Assume Schinzel's hypothesis holds. Then the Brauer–Manin obstruction to the Hasse principle and weak approximation for rational points is the only obstruction for any smooth proper model of V.*

In Section 4, we prove the following theorem.

THEOREM 1.2. *Let V be the smooth locus of one of the above three varieties. If there is no Brauer–Manin obstruction to the existence of a zero-cycle of degree 1 on a smooth proper model V^c of V, then there is a zero-cycle of degree 1 on V^c (defined over k).*

2. *Some recollections from* [8] *and some complements*

Let us recall some results from [8]. Let k be a field of characteristic zero, $\bar{k}$ a fixed algebraic closure of k, Γ_k the absolute Galois group of k. Let $P(t) \in k[t]$ be a non-zero polynomial. We consider the affine variety over k defined by

$$N_{K/k}(\Xi) = P(t), \tag{2.1}$$

where K/k is *a finite product of finite separable field extensions of* k, Ξ is a 'variable' in K and $N_{K/k}$ is the formal norm associated with K/k. In [8], K/k is a field extension, in fact, much of the theory holds for K/k a product of fields. This observation is used in subsequent sections (for example, Lemma 3.5 and Theorem 4.5).

Let $V \subset \mathbb{A}^{n+1}_k \simeq \mathbb{A}^1_k \times R_{K/k}(\mathbb{A}^1_K)$ be the *smooth locus* of the affine hypersurface defined by equation (2.1). If $P(t)$ is separable, then V is the hypersurface defined by (2.1). The projection $(\Xi, t) \mapsto t$ defines a surjective morphism $p : V \to \mathbb{A}^1_k$. Let $U_0 \subset \mathbb{A}^1_k$ be the open subset defined by $P(t) \neq 0$ and $U = p^{-1}(U_0) \subset V$.

Let $T = R^1_{K/k}(\mathbb{G}_m)$, $\hat{T}$ the Γ_k-module of characters of T and T^c a smooth T-equivariant compactification (see [9]). The contracted product $U \times^T T^c$ is a partial compactification of U which is proper and smooth over U_0. Let X be the k-variety by gluing V and $U \times^T T^c$ along U, which is separated and smooth over k. In this paper, X is called the *CTHS* partial compactification of V, which is first constructed in [8].

For any k-variety V, we denote $\bar{V} := V \times_k \bar{k}$. In [8, Section 2], the following results are established: The natural map $\bar{k}^\times \to \bar{k}[X]^\times$ is an isomorphism, $\mathrm{Pic}(\bar{X})$ is finitely generated and torsion-free, and $\mathrm{Br}(\bar{X}) = 0$.

By Hironaka's theorem [24], one may find a smooth proper compactification X^c which contains X with a morphism $X^c \to \mathbb{P}^1_k$ extending the morphism $X \to \mathbb{A}^1_k$. There are natural inclusions

$$\mathrm{Br}(X^c) \hookrightarrow \mathrm{Br}(X).$$

One is interested in knowing when these are equalities.

There are also natural inclusions

$$\mathrm{Br}_{\mathrm{vert}}(X) \hookrightarrow \mathrm{Br}(X)$$

and

$$\mathrm{Br}_{\mathrm{vert}}(X^c) \hookrightarrow \mathrm{Br}_{\mathrm{vert}}(X^c),$$

where $\mathrm{Br}_{\mathrm{vert}}(X)$ denotes the subgroup of $\mathrm{Br}(X)$ whose image in the Brauer group $\mathrm{Br}(X_\eta)$ of the generic fibre of $X \to \mathbb{P}^1_k$ is in the image of $\mathrm{Br}(k(\mathbb{P}^1)) \to \mathrm{Br}(X_\eta)$, and similarly for $\mathrm{Br}_{\mathrm{vert}}(X^c)$. One is interested in deciding which groups contribute to $\mathrm{Br}(X)$ and to $\mathrm{Br}(X^c)$, and ultimately in computing $\mathrm{Br}(X^c)$.

Let F/k be a Galois extension and M a $\mathrm{Gal}(F/k)$-module. For simplicity, in this paper, we always denote

$$H^i(F/k, M) := H^i(\mathrm{Gal}(F/k), M).$$

If $F = \bar{k}$, then we denote it simply by $H^i(k, M)$. We define

$$Ш^i_\omega(F/k, M) := \bigcap_{g\in \mathrm{Gal}(F/k)} \mathrm{Ker}[H^i(F/k, M)\to H^i(\langle g\rangle, M)],$$

where $\langle g\rangle$ is the cyclic subgroup of $\mathrm{Gal}(F/k)$ generated by g. If $F = \bar{k}$, then we also simply denote $Ш^i_\omega(\bar{k}/k, M)$ by $Ш^i_\omega(M)$.

We denote $\mathbb{Z}[L/k]$ to be the Γ_k-module $\mathbb{Z}[\Gamma_k/\Gamma_L]$ for any finite field extension L/k. Write $P(t) = cp_1(t)^{e_1}\cdots p_m(t)^{e_m}$, where $p_i(t)$ is monic and irreducible. Denote by $\mathbb{Z}_P$ the permutation Γ_k-module associated with the polynomial $P(t)$, which is a direct sum of $\mathbb{Z}[L_i/k]$, where $L_i = k[t]/(p_i(t))$. Let $N_i = N_{L_i/k} \in \mathbb{Z}[L_i/k]$ be $\sum_\sigma \sigma$ with σ running over all embeddings of L_i into $\bar{k}$.

Let $j_P : \mathbb{Z} \to \mathbb{Z}_P$ be defined by sending 1 to $-(e_1N_1 + \cdots + e_mN_m)$. Let F/k be a Galois extension which splits $P(t)$. For any $\mathrm{Gal}(F/k)$-module M, j_P induces a morphism $M \to M \otimes \mathbb{Z}_P$. We define

$$Ш^2_\omega(F/k, M)_P := \mathrm{Ker}[Ш^2_\omega(F/k, M)\to Ш^2_\omega(F/k, M \otimes \mathbb{Z}_P)].$$

If $F = \bar{k}$, then we simply denote it by $Ш^2_\omega(M)_P$.

With the notation as above, Colliot-Thélène, Harari and Skorobogatov proved the following theorem.

THEOREM 2.1 ([8, Proposition 2.5]). *Let X/k be the CTHS partial compactification of V as above. Then*:

(i) *the following sequence is exact*

$$0 \longrightarrow H^1(k, \hat{T} \otimes \mathbb{Z}_P)/j_{P*}H^1(k, \hat{T}) \longrightarrow H^1(k, \mathrm{Pic}(\bar{X})) \longrightarrow Ш^2_\omega(\hat{T})_P \longrightarrow 0; \tag{2.2}$$

(ii) *the elements of* $\mathrm{Br}(X)$ *whose image in* $H^1(k, \mathrm{Pic}(\bar{X}))$ *come from* $H^1(k, \hat{T} \otimes \mathbb{Z}_P)$ *are precisely the elements of* $\mathrm{Br}_{\mathrm{vert}}(X)$.

Under the assumption $H^3(k, \bar{k}^\times) = 0$, which is satisfied if k is a number field, then the above exact sequence (2.2) identifies with

$$0 \longrightarrow \mathrm{Br}_{\mathrm{vert}}(X)/\mathrm{Br}_0(X) \longrightarrow \mathrm{Br}(X)/\mathrm{Br}_0(X) \longrightarrow Ш^2_\omega(\hat{T})_P \longrightarrow 0, \tag{2.3}$$

where $\mathrm{Br}_0(X)$ denotes the image of the natural map $\mathrm{Br}(k)\to\mathrm{Br}(X)$.

Let us mention an easy application of Theorem 2.1.

PROPOSITION 2.2. *Let $k, K, P(t), X$ be as above. Assume that K is a field, and that the extension K/k is abelian, and that $P(t)$ is irreducible. Then*

(a) $H^1(k, \hat{T} \otimes \mathbb{Z}_P)/j_{P*}H^1(k, \hat{T}) = 0$;
(b) $\mathrm{Br}_{\mathrm{vert}}(X) \simeq \mathrm{Br}_0(X)$;
(c) *if* $H^3(k, \bar{k}^\times) = 0$, *then* $\mathrm{Br}(X)/\mathrm{Br}_0(X) \simeq Ш^2_\omega(\hat{T})_P$;
(d) *if K/k is cyclic, then* $H^1(k, \mathrm{Pic}(\bar{X})) = 0$, *hence* $\mathrm{Br}(X) = \mathrm{Br}_0(X)$.

Proof. Let $L = k[t]/(P(t))$. We know

$$\begin{aligned} H^1(k, \hat{T} \otimes \mathbb{Z}_P) &\simeq H^1(L, \hat{T}) \simeq \mathrm{Ker}[H^1(L, \mathbb{Q}/\mathbb{Z})\to H^1(L.K, \mathbb{Q}/\mathbb{Z})] \\ &\simeq \mathrm{Hom}(\mathrm{Gal}(L.K/L), \mathbb{Q}/\mathbb{Z}). \end{aligned}$$

Since the extension K/k is abelian, the restriction map

$$H^1(K/k, \mathbb{Q}/\mathbb{Z}) \longrightarrow H^1(L.K/L, \mathbb{Q}/\mathbb{Z}) \text{ is surjective,}$$

hence $H^1(k, \hat{T} \otimes \mathbb{Z}_P)/j_{P*}H^1(k, \hat{T}) = 0$, (b) and (c) follows from the sequences (2.2) and (2.3).

If K/k is cyclic, since $\hat{T}$ is split by K, then

$$Ш^2_\omega(\hat{T}) = Ш^2_\omega(K/k, \hat{T}) = \bigcap_{g \in \mathrm{Gal}(K/k)} \mathrm{Ker}[H^2(K/k, M) \to H^2(\langle g \rangle, M)] = 0.$$

By the sequence (2.2), we have $H^1(k, \mathrm{Pic}(\bar{X})) = 0$. □

Proposition 2.3. *Let $k, K, P(t), L_i, X$ and X^c be as above. The natural surjective map $\mathrm{Pic}(\bar{X}^c) \to \mathrm{Pic}(\bar{X})$ induces an injection*

$$H^1(k, \mathrm{Pic}(\bar{X}^c)) \hookrightarrow H^1(k, \mathrm{Pic}(\bar{X}))$$

and isomorphisms

$$Ш^1_\omega(\mathrm{Pic}(\bar{X}^c)) \simeq Ш^1_\omega(\mathrm{Pic}(\bar{X})) \simeq Ш^2_\omega(\hat{T}'),$$

where T' is the torus over k defined by

$$\prod_{i=1}^{m} N_{L_i/k}(\Upsilon_i)^{e_i} \cdot N_{K/k}(\Xi) = 1.$$

Proof. Let $\mathrm{Div}_{\bar{X}^c \setminus \bar{X}}(\bar{X}^c)$ be the group of divisors of $\bar{X}^c$ with supports in $\bar{X}^c \setminus \bar{X}$. Since $\bar{k}[X]^\times = \bar{k}^\times$, we have the following exact sequence:

$$0 \longrightarrow \mathrm{Div}_{\bar{X}^c \setminus \bar{X}}(\bar{X}^c) \longrightarrow \mathrm{Pic}(\bar{X}^c) \longrightarrow \mathrm{Pic}(\bar{X}) \longrightarrow 0,$$

then the injectivity of

$$H^1(k, \mathrm{Pic}(\bar{X}^c)) \longrightarrow H^1(k, \mathrm{Pic}(\bar{X}))$$

is implied by $H^1(k, \mathrm{Div}_{\bar{X}^c \setminus \bar{X}}(\bar{X}^c)) = 0$, which is obvious since $\mathrm{Div}_{\bar{X}^c \setminus \bar{U}}(\bar{X}^c)$ is a permutation Γ_k-module.

Let $U (\subset X \subset X^c)$ be the smooth affine variety defined by

$$N_{K/k}(\Xi) = P(t) \neq 0.$$

Let $\bar{U}_0 \subset \mathbb{A}^1_k$ be a open subset defined $P(t) \neq 0$. There exists an isomorphism of varieties over $\bar{k}$

$$\bar{U} \simeq \bar{U}_0 \times \bar{T} \simeq \bar{U}_0 \times \mathbb{G}_m^{[K:k]-1}, \tag{2.4}$$

hence $\mathrm{Pic}(\bar{U}) = 0$. Therefore, we have the following exact sequence:

$$0 \longrightarrow \bar{k}[U]^\times/\bar{k}^\times \longrightarrow \mathrm{Div}_{\bar{X}^c \setminus \bar{U}}(\bar{X}^c) \longrightarrow \mathrm{Pic}(\bar{X}^c) \longrightarrow 0.$$

Since $\mathrm{Div}_{\bar{X}^c \setminus \bar{U}}(\bar{X}^c)$ is a permutation Γ_k-module, we have $H^1(k, \mathrm{Div}_{\bar{X}^c \setminus \bar{U}}(\bar{X}^c)) = 0$. Then we have the following exact sequence:

$$0 \longrightarrow H^1(k, \mathrm{Pic}(\bar{X}^c)) \longrightarrow H^2(k, \bar{k}[U]^\times/\bar{k}^\times) \longrightarrow H^2(k, \mathrm{Div}_{\bar{X}^c \setminus \bar{U}}(\bar{X}^c)).$$

Thus, we obtain the exact sequence

$$0 \longrightarrow Ш^1_\omega(\mathrm{Pic}(\bar{X}^c)) \longrightarrow Ш^2_\omega(\bar{k}[U]^\times/\bar{k}^\times) \longrightarrow Ш^2_\omega(\mathrm{Div}_{\bar{X}^c \setminus \bar{U}}(\bar{X}^c)).$$

Since $\mathrm{Div}_{\bar{X}^c \setminus \bar{U}}(\bar{X}^c)$ is a permutation Γ_k-module, we have

$$Ш^2_\omega(\mathrm{Div}_{\bar{X}^c \setminus \bar{U}}(\bar{X}^c)) = 0.$$

Thus $Ш^1_\omega(\mathrm{Pic}(\bar{X}^c)) \simeq Ш^2_\omega(\bar{k}[U]^\times/\bar{k}^\times)$.

Since $\bar{k}[X]^\times = \bar{k}^\times$, we also have the exact sequence

$$0 \longrightarrow \bar{k}[U]^\times/\bar{k}^\times \longrightarrow \mathrm{Div}_{\bar{X}\setminus\bar{U}}(\bar{X}) \longrightarrow \mathrm{Pic}(\bar{X}) \longrightarrow 0.$$

By similar arguments as above, we also have $Ш^1_\omega(\mathrm{Pic}(\bar{X})) \simeq Ш^2_\omega(\bar{k}[U]^\times/\bar{k}^\times)$.

By (2.4), we know $\bar{k}[U]^\times/\bar{k}^\times$ is generated by the linear factors of $P(t)$ over $\bar{k}$ and all regular functions over $\bar{k}$ of the torus $R_{K/k}(\mathbb{G}_m)$, hence $\bar{k}[U]^\times/\bar{k}^\times \simeq \hat{T}'$ as Γ_k-modules, which implies $Ш^2_\omega(\bar{k}[U]^\times/\bar{k}^\times) \simeq Ш^2_\omega(\hat{T}')$. □

3. *Calculation of the Brauer group*

In the remainder of this paper, we always assume K/k is a field extension. In this section, we keep notations as in [8] and as in Section 1. In [8], Colliot-Thélène, Harari and Skorobogatov gave a formula for the vertical Brauer group $\mathrm{Br}_{\mathrm{vert}}(X)$ and the quotient $\mathrm{Br}(X)/\mathrm{Br}_{\mathrm{vert}}(X)$. They pointed out that the vertical unramified Brauer group $\mathrm{Br}_{\mathrm{vert}}(X^c)$ can be calculated by the formula. However, it is still open how to determine the unramified Brauer group $\mathrm{Br}(X^c)$. The aim of this section is to investigate the latter group.

3.1. *The case $P(t)$ is irreducible and some linear independence condition is satisfied*

LEMMA 3.1. *Let $P(t)$ be an irreducible polynomial and $L = k[t]/(P(t))$. Let K^{cl} (respectively, L^{cl}) be the Galois closure of K (respectively, L) over k. If $L \cap K^{cl} = k$, then $Ш^2_\omega(\hat{T})_P = 0$.*

Proof. Let E be the compositum $K^{cl}.L^{cl}$. We can see $Ш^2_\omega(\hat{T}) = Ш^2_\omega(E/k, \hat{T})$ since $\hat{T}$ is split by E. Then

$$\begin{aligned} Ш^2_\omega(\hat{T})_P &= \mathrm{Ker}[j_{P*} : Ш^2_\omega(\hat{T}) \to Ш^2_\omega(\hat{T} \otimes \mathbb{Z}_P)] \\ &= \mathrm{Ker}[j_{P*} : Ш^2_\omega(E/k, \hat{T}) \to Ш^2_\omega(E/k, \hat{T} \otimes \mathbb{Z}_P)] \\ &= \mathrm{Ker}[j_{P*} : Ш^2_\omega(E/k, \hat{T}) \to H^2(E/k, \hat{T} \otimes \mathbb{Z}_P)]. \end{aligned}$$

Using Shapiro's lemma, we have

$$Ш^2_\omega(\hat{T})_P = \mathrm{Ker}[\mathrm{Res}_{k/L} : Ш^2_\omega(E/k, \hat{T}) \to H^2(E/L, \hat{T})].$$

We have the following commutative diagram:

$$\begin{array}{ccc} Ш^2_\omega(E/k, \hat{T}) & \longrightarrow & H^2(E/L, \hat{T}) \\ {\scriptstyle f}\uparrow & & {\scriptstyle g}\uparrow \\ Ш^2_\omega(K^{cl}/k, \hat{T}) & \rightarrowtail & H^2(L.K^{cl}/L, \hat{T}) \end{array}$$

Since $\hat{T}$ is also split by K^{cl}, f is an isomorphism. By the Hochschild–Serre spectral sequence, one obtains the exact sequence

$$H^1(E/L.K^{cl}, \hat{T})^{\mathrm{Gal}(E/L)} \longrightarrow H^2(L.K^{cl}/L, \hat{T}) \longrightarrow H^2(E/L, \hat{T}).$$

Obviously, $H^1(E/L.K^{cl}, \hat{T}) = 0$, hence g is injective. So we have

$$Ш^2_\omega(\hat{T})_P = \mathrm{Ker}[Ш^2_\omega(K^{cl}/k, \hat{T}) \to H^2(L.K^{cl}/L, \hat{T})].$$

Since $L \cap K^{cl} = k$, we have

$$H^2(K^{cl}/k, \hat{T}) \simeq H^2(L.K^{cl}/L, \hat{T}),$$

hence $Ш^2_\omega(\hat{T})_P = 0$. □

THEOREM 3.2. *Let $P(t)$ be an irreducible polynomial over k and $L = k[t]/(P(t))$. Let K^{cl} be the Galois closure of the field extension K/k. Let X/k be the CTHS partial compactification for $P(t)$ and K as in Section* 1.

If $L \cap K^{cl} = k$, then $H^1(k, \mathrm{Pic}(\bar{X})) = 0$, hence $\mathrm{Br}(X) = \mathrm{Br}_0(X)$.

Proof. By the injectivity of $\mathrm{Br}(X)/\mathrm{Br}_0(X) \to H^1(k, \mathrm{Pic}(\bar{X}))$, we only need to show $H^1(k, \mathrm{Pic}(\bar{X})) = 0$. Since $L \cap K^{cl} = k$, we have $Ш^2_\omega(\hat{T})_P = 0$ by Lemma 3.1. By the sequence (2.2), we only need to show $H^1(k, \hat{T} \otimes \mathbb{Z}_P)/j_{P*}H^1(k, \hat{T}) = 0$.

By Shapiro's lemma and the assumption $L \cap K^{cl} = k$, we have the following commutative diagram:

$$\begin{array}{ccc} H^1(k,\hat{T}) & \xrightarrow{j_{P*}} & H^1(k,\hat{T}\otimes\mathbb{Z}_P) \\ & \searrow^{\mathrm{Res}_{k/L}} & \downarrow\cong \\ \downarrow\cong & & H^1(L,\hat{T}) \\ & & \downarrow\cong \\ H^1(K^{cl}/k,\hat{T}) & \xrightarrow{\cong} & H^1(L.K^{cl}/L,\hat{T}) \end{array}$$

which implies

$$H^1(E/k, \hat{T} \otimes \mathbb{Z}_P)/j_{P*}H^1(E/k, \hat{T}) = 0. \qquad \square$$

By the above theorem and Proposition 2.3, we have an application for certain multi-norm tori.

COROLLARY 3.3. *Let L and K be field extensions over k. Let T' be the torus over k defined by $N_{L/k}(\Upsilon)N_{K/k}(\Xi) = 1$. If $L \cap K^{cl} = k$ or K/k is cyclic, then $Ш^2_\omega(\hat{T'}) = 0$, in particular, if k is a number field, then principal homogeneous spaces of T' satisfy the Hasse principle and weak approximation.*

Proof. Let $P(t)$ be an irreducible polynomial over k such that $L \simeq k[t]/(P(t))$. Let X be the *CTHS* partial compactification (see Section 1) for $P(t)$ and K. By Proposition 2.3, we have $Ш^2_\omega(\hat{T'}) \subset H^1(k, \mathrm{Pic}(\bar{X}))$. The statement follows from Proposition 2.2 for K/k cyclic and Theorem 3.2 for $L \cap K^{cl} = k$.

Suppose that k is a number field. Since $Ш^2_\omega(\hat{T'}) = 0$, the result follows from the fact that the Brauer–Manin obstruction is the only one to the Hasse principle and weak approximation for principal homogeneous spaces of tori [**35**, Theorem 8.12] or [**37**, Theorem 5.2.1]. $\square$

REMARK 3.4. The case that K/k is cyclic in Corollary 3.3 was proved in an unpublished paper of Sansuc. If $L^{cl} \cap K^{cl} = k$, then this corollary was also proved by Pollio and Rapinchuk (see [**32**]), and a more general result was proved in [**19**].

3.2. *The case $P(t)$ irreducible and K/k an abelian field extension*

In this section, we investigate the case that $P(t)$ is irreducible and K/k abelian. The main result is Theorem 3.6, which, in many situations, gives good control of the quotient $\mathrm{Br}(X)/\mathrm{Br}(X^c)$.

LEMMA 3.5. *Let $P(t)$ be an irreducible polynomial over k, K/k an abelian extension, and X the CTHS partial compactification in Section 1. Denote by K^g the fixed field of K by $g \in \Gamma_k$, $L = k[t]/(P(t))$ and $L_g = L \cap K^g$.*

Then any element of $\mathrm{Br}(X_{K^g})$ which comes from $\mathrm{Br}(X)$ has the form

$$\rho + \sum_{\sigma \in \mathrm{Gal}(L_g/k)} \psi(\sigma)(p(t)^\sigma, \chi),$$

where $\rho \in \mathrm{Br}(K^g)$, $p(t)$ is a fixed irreducible factor of $P(t)$ over L_g, χ is a fixed primitive character of $\mathrm{Gal}(K/K^g)$, $\psi \in \mathrm{Hom}(\mathrm{Gal}(L_g/k), \mathbb{Z}/b\mathbb{Z})$ with $b := [K : K \cap (L.K^g)]$.

Proof. The proof is based on Theorem 2.1, and the observation that $\mathrm{Br}(X_{K^g}) = Br_{\mathrm{vert}}(X_{K^g})$. Since K/K^g is cyclic, we have

$$Ш^2_\omega(K^g, \hat{T}) = Ш^2_\omega(K/K^g, \hat{T}) = 0.$$

By the sequence (2.2), we have the commutative diagram:

$$\begin{array}{ccccc} 0 & \longrightarrow & H^1(k, \hat{T} \otimes \mathbb{Z}_P)/j_{P*}H^1(k, \hat{T}) & \longrightarrow & H^1(k, \mathrm{Pic}(\bar{X})) \\ & & \downarrow & & \downarrow \mathrm{Res} \\ 0 & \longrightarrow & H^1(K^g, \hat{T} \otimes \mathbb{Z}_P)/j_{P*}H^1(K^g, \hat{T}) & \xrightarrow{\cong} & H^1(K^g, \mathrm{Pic}(\bar{X})) \end{array}$$

We want to investigate the restriction of $\mathrm{Br}(X)$ to $\mathrm{Br}(X_{K^g})$, that is, the restriction of $H^1(k, \mathrm{Pic}(\bar{X}))$ to $H^1(K^g, \mathrm{Pic}(\bar{X}))$. Since the image of $H^1(k, \mathrm{Pic}(\bar{X}))$ is Γ_k-invariant, we only need consider $[H^1(K^g, \hat{T} \otimes \mathbb{Z}_P)/j_{P*}H^1(K^g, \hat{T})]^{\Gamma_k/\Gamma_{K^g}}$ by the above commutative diagram, noting that Γ_{K^g} is a normal subgroup of Γ_k since K/k is abelian.

We recall $L_g = L \cap K^g$. Write $P(t) = cp_1(t) \cdots p_m(t)$, where $c \in k^\times$ and $p_i(t) \in L_g[t]$ monic and irreducible. Since $L_g \subset K$ and K/k abelian, L_g/k is Galois (abelian). Denote $\Delta := \mathrm{Gal}(L_g/k)$, then Δ acts transitively on $\{p_i(t) : 1 \leqslant i \leqslant m\}$. We fix $p(t)$ to be some $p_i(t)$ and let $L' = L_g[t]/(p(t))$. Therefore, $L \cong L'$ over k and

$$\mathbb{Z}_P \simeq \mathbb{Z}[L.K^g/K^g] \otimes \mathbb{Z}[\Delta]$$

as Γ_k-modules, which implies the natural isomorphism

$$H^1(K^g, \hat{T} \otimes \mathbb{Z}_P) \simeq H^1(K^g, \hat{T} \otimes \mathbb{Z}[L.K^g/K^g]) \otimes \mathbb{Z}[\Delta] \simeq H^2(L.K/L.K^g, \mathbb{Z}) \otimes \mathbb{Z}[\Delta].$$

Denote

$$R := H^1(L.K/L.K^g, \mathbb{Q}/\mathbb{Z}) \simeq H^1(K/K \cap (L.K^g), \mathbb{Q}/\mathbb{Z}), \tag{3.1}$$

then $H^1(K^g, \hat{T} \otimes \mathbb{Z}_P) \simeq R[\Delta]$.

The following diagram

$$\begin{array}{ccc} H^1(K^g, \hat{T}) & \xrightarrow{j_{P*}} & H^1(K^g, \hat{T} \otimes \mathbb{Z}_P) \\ \downarrow \cong & & \downarrow \cong \\ H^1(K/K^g, \mathbb{Q}/\mathbb{Z}) & \xrightarrow{\mathrm{Res}} & R[\Delta] \end{array}$$

is commutative, where the map Res is the diagonal map by restriction to R. Since K/k is abelian, the image of Res in $R[\Delta]$ is $R \cdot N$ and R is a constant Γ_k-module, where $N = \sum_{\sigma \in \Delta} \sigma \in R[\Delta]$. Then we have an isomorphism as Γ_k-modules

$$H^1(K^g, \hat{T} \otimes \mathbb{Z}_P)/j_{P*}H^1(K^g, \hat{T}) \simeq R[\Delta]/(R \cdot N).$$

Then

$$[H^1(K^g, \hat{T} \otimes \mathbb{Z}_P)/j_{P*}H^1(K^g, \hat{T})]^{\Gamma_k/\Gamma_{K^g}} \simeq (R[\Delta]/(R \cdot N))^\Delta.$$

The long exact sequence in Galois cohomology associated to the exact sequence

$$0 \longrightarrow R \longrightarrow R[\Delta] \longrightarrow R[\Delta]/(R \cdot N) \longrightarrow 0$$

gives the exact sequence

$$0 \longrightarrow R \xrightarrow{\cong} R \cdot N \longrightarrow (R[\Delta]/(R \cdot N))^{\Delta} \longrightarrow \mathrm{Hom}(\Delta, R) \longrightarrow 0.$$

Therefore, $(R[\Delta]/(R \cdot N))^{\Delta} \simeq \mathrm{Hom}(\Delta, R)$. We now describe this isomorphism more explicitly.

Let $\psi \in \mathrm{Hom}(\Delta, R)$, we define

$$f(\psi) := \sum_{\sigma \in \Delta} \psi(\sigma)\sigma \in R[\Delta]$$

and $\bar{f}(\psi)$ is the image of $f(\psi)$ in $R[\Delta]/R \cdot N$. Let $\sigma_0 \in \Delta$, we have

$$\begin{aligned} \sigma_0(f(\psi)) - f(\psi) &= \sum_{\sigma} \psi(\sigma)\sigma_0\sigma - \sum_{\sigma} \psi(\sigma)\sigma \\ &= \sum_{\sigma} (\psi(\sigma) - \psi(\sigma_0))\sigma - \sum_{\sigma} \psi(\sigma)\sigma \\ &= -\psi(\sigma_0) \sum_{\sigma} \sigma \in R \cdot N, \end{aligned}$$

hence $\bar{f}(\psi) \in (R[\Delta]/(R \cdot N))^{\Delta}$. On the other hand, we suppose $\bar{f}(\psi_1) = \bar{f}(\psi_2)$. Then $\sum_{\sigma}(\psi_1(\sigma) - \psi_2(\sigma))\sigma \in R \cdot N$, that is, $\psi_1(\sigma) - \psi_2(\sigma)$ is a constant for all $\sigma \in \Delta$. Let e be the identity of Δ, we have $\psi_1(e) - \psi_2(e) = 0$. Then $\psi_1 = \psi_2$. Therefore, we have

$$[R[\Delta]/(R \cdot N)]^{\Delta} = \{\bar{f}(\psi) \in R[\Delta]/R \cdot N : \psi \in \mathrm{Hom}(\Delta, R)\}. \tag{3.2}$$

Since K/K^g is cyclic, we can choose a primitive character χ of $\mathrm{Gal}(K/K^g)$, then $\mathrm{Res}(\chi)$ is also a primitive character of $\mathrm{Gal}(L.K/L.K^g)$, that is, a generator of R (see (3.1) for the definition of R). Recalled $b := [K : K \cap (L.K^g)] = \#R$. Rewrite equation (3.2), we have

$$[R[\Delta]/(R \cdot N)]^{\Delta} = \{\bar{f}'(\psi) \in R[\Delta]/R \cdot N : \psi \in \mathrm{Hom}(\Delta, \mathbb{Z}/b\mathbb{Z})\}, \tag{3.3}$$

where $f'(\psi) := \sum_{\sigma \in \Delta} \psi(\sigma)\mathrm{Res}(\chi) \cdot \sigma \in R[\Delta]$.

For any field k with $char(k) = 0$, using Grothendieck's purity theorem (cf. [**18**, Theorem 1.3.2]), we can give a precise description of the map $\mathrm{Br}(X) \to H^1(k, \mathrm{Pic}(\bar{X}))$ by the observation 3 of [**28**, Section 2]. Then we have the following commutative diagram:

$$\begin{array}{ccc} \mathrm{Br}_{\mathrm{vert}}(X_{K^g}) & \xrightarrow{\phi} & R[\Delta] \\ & \searrow & \downarrow \\ & & H^1(K_g, \mathrm{Pic}(\bar{X})) \end{array}$$

where ϕ is induced by the residue map of $\mathrm{Br}(K^g(t))$. Using Faddeev's sequence (cf. [**18**, Section 1.2]), equation (3.3) implies that any element of $\mathrm{Br}(X_{K^g})$ $(= \mathrm{Br}_{\mathrm{vert}}(X_{K^g}))$ which comes from $\mathrm{Br}(X)$ has the form

$$\begin{aligned} &\rho + \sum_{\sigma \in \mathrm{Gal}(L_g/k)} \psi(\sigma)\mathrm{Cor}_{L^{\sigma}/k}(t - \sigma(\eta), \mathrm{Res}(\chi)) \\ &= \rho + \sum_{\sigma \in \mathrm{Gal}(L_g/k)} \psi(\sigma)(p(t)^{\sigma}, \chi), \end{aligned}$$

where $\rho \in \mathrm{Br}(K^g)$ and η is a root of $p(t)$ in $\bar{k}^{\times}$. This argument is similar as in the remark of [**8**, p. 76]. □

THEOREM 3.6. *Suppose* $\mathrm{char}(k) = 0$. *Let* $P(t)$ *be an irreducible polynomial over* k, K/k *an abelian extension and* $L = k[t]/(P(t))$. *Let* X *be the CTHS partial compactification for* $P(t)$ *and* K *as in Section* 1, X^c *a smooth compactification of* X. *Then*:

(a) *the quotient* $\mathrm{Br}(X)/\mathrm{Br}(X^c)$ *is 2-torsion*;
(b) $\mathrm{Br}(X^c) = \mathrm{Br}(X)$ *if one of the following conditions holds*:
  (1) $\mathrm{Gal}(K/k) \simeq \mathbb{Z}/2^i \times A$, *where* $i \geqslant 0$ *and* A *has odd order*;
  (2) $[L \cap K : k]$ *is odd*;
  (3) $[L : L \cap K]$ *is even*;
  (4) *there exists* $s \geqslant 1$ *such that* $2^s \mid [L : k]$ *and the cokernel of the multiplication by* 2^{s-1} *on* $\mathrm{Gal}(K/k)$ *has odd order*;
  (5) L/k *contains an abelian subfield* L'/k *with* $\mathrm{Gal}(L'/k) \cong (\mathbb{Z}/2\mathbb{Z})^3$.

Proof. Let $k(X)$ be the function field of X. For each discrete valuation ring A which contains k and with fraction field $k(X)$ and residue field κ_A, there is a residue map

$$\partial_A : \mathrm{Br}(k(X)) \longrightarrow H^1(\kappa_A, \mathbb{Q}/\mathbb{Z}).$$

Since $char(k) = 0$, Grothendieck's purity theorem (cf. [18, Theorem 1.3.2]) gives

$$\mathrm{Br}(X^c) = \bigcap_A \mathrm{Ker}(\partial_A) \subset \mathrm{Br}(k(X)),$$

where A runs through all discrete valuation ring as above.

Let $\mathcal{B} \in \mathrm{Br}(X)$, first we will prove $2\mathcal{B} \in \mathrm{Br}(X^c)$, that is, for any $g \in \mathrm{Gal}(\bar{\kappa}_A/\kappa_A)$,

$$2\partial_A(\mathcal{B})(g) = 0 \in \mathbb{Q}/\mathbb{Z}$$

by Grothendieck's purity theorem.

Since $k \subset \kappa_A$, we can fix an embedding $\bar{k} \hookrightarrow \bar{\kappa}_A$. Let K^g be the fixed field of K by g and $L_g = L \cap K^g$. Let f be the natural map $\mathrm{Br}(X) \to \mathrm{Br}(X_{K^g})$. By Lemma 3.5, there exists $\psi_{\mathcal{B}} \in \mathrm{Hom}(\mathrm{Gal}(L_g/k), \mathbb{Z}/b)$ such that

$$f(\mathcal{B}) = \rho + \sum_{\sigma \in \mathrm{Gal}(L_g/k)} \psi_{\mathcal{B}}(\sigma)(p(t)^\sigma, \chi), \tag{3.4}$$

where $\rho \in \mathrm{Br}(K^g)$, $b = [K : K \cap (L.K^g)]$, $p(t)$ is a fix irreducible factor of $P(t)$ over L_g, and χ is a fixed primitive character of $\mathrm{Gal}(K/K^g)$.

Since X is geometrically integral, the function field $K(X) \cong k(X) \otimes_k K$ of $X \times_k K$ is finite and unramified over $k(X)$. We can extend A to a discrete valuation ring A_K of $K(X)$ with residue field $\kappa_{A_K} = \kappa_A.K$. Indeed, the completion of $k(X)$ for the given valuation is isomorphic to $\kappa_A((\pi))$, where π is a uniformizer. Considering the valuation given by π on $(K.\kappa_A)((\pi))$ and using $K(X) = k(X) \otimes_k K$, since $K(X)$ can naturally embed into $(K.\kappa_A)((\pi))$ and its image is dense in the latter, there is a discrete valuation ring $A_K \subset K(X)$ and its residue field is just $K.\kappa_A$.

For any intermediate field M of K/k, we have similarly the valuation ring A_M of $M(X)$ with residue field κ_{A_M}. We write A_g for A_{K^g}.

By [18, Proposition 1.1.1], we have the commutative diagram:

$$\begin{array}{ccc} \mathrm{Br}(k(X)) & \xrightarrow{\partial_A} & H^1(\kappa_A, \mathbb{Q}/\mathbb{Z}) \\ {\scriptstyle \mathrm{Res}_{k/K^g}}\downarrow & & \downarrow{\scriptstyle \mathrm{Res}_{\kappa_A/\kappa_{A_g}}} \\ \mathrm{Br}(K^g(X)) & \xrightarrow{\partial_{A_g}} & H^1(\kappa_{A_g}, \mathbb{Q}/\mathbb{Z}) \end{array}$$

Since $\kappa_{A_g} = \kappa_A.K^g$, we have $g \in \mathrm{Gal}(\bar{\kappa}_A/\kappa_{A_g})$. Hence

$$\partial_A(\mathcal{B})(g) = \partial_{A_g}(f(\mathcal{B}))(g).$$

Therefore, we only need to show $2\partial_{A_g}(f(\mathcal{B})) = 0$.

We consider two cases.

(i) The case $\mathrm{ord}_{A_g}(t) \geqslant 0$.

If $\mathrm{ord}_{A_g}(p(t)^\sigma) = 0$ for all $\sigma \in \mathrm{Gal}(L_g/k)$, then it is clear $\partial_{A_g}(f(\mathcal{B})) = 0$ by (3.4). If there is a $\sigma_0 \in \mathrm{Gal}(L_g/k)$ such that $\mathrm{ord}_{A_g}(p_1(t)^{\sigma_0}) > 0$. It is clear that

$$\mathrm{ord}_{A_g}(p_1(t)^\sigma) = 0 \quad \text{for } \sigma \neq \sigma_0$$

since $p_1(t)^\sigma$ and $p_1(t)^{\sigma_0}$ are relatively prime. Then we have

$$\mathrm{ord}_{A_g}(p_1(t)^{\sigma_0}) = \mathrm{ord}_{A_g}(P(t)).$$

Therefore,

$$\partial_{A_g}(f(\mathcal{B})) = \psi_{\mathcal{B}}(\sigma_0)\,\mathrm{ord}_{A_g}(p_1(t)^{\sigma_0}) \cdot \bar{\chi} = \psi_{\mathcal{B}}(\sigma_0)\mathrm{ord}_{A_g}(P(t)) \cdot \bar{\chi} = 0$$

by the equation $N_{K/k}(\Xi) = P(t)$.

(ii) The case $\mathrm{ord}_{A_g}(t) = \delta < 0$.

Denote $d := \deg(p(t))$. Then

$$\mathrm{ord}_{A_g}(p(t)^\sigma) = \delta \cdot \deg(p(t)^\sigma) = \delta d.$$

Therefore,

$$\partial_{A_g}(f(\mathcal{B})) = \sum_\sigma \psi_{\mathcal{B}}(\sigma)\,\mathrm{ord}_{A_g}(p(t)^\sigma) \cdot \bar{\chi} = \delta d \sum_\sigma \psi_{\mathcal{B}}(\sigma) \cdot \bar{\chi}.$$

Denote $\psi_{\mathcal{B}}(\mathrm{Gal}(L_g/k)) := \langle u \rangle \subset \mathbb{Z}/b$ with $u \mid b$. Then we have

$$\begin{aligned}\partial_{A_g}(f(\mathcal{B})) &= \delta d \cdot \#\mathrm{Ker}(\psi_{\mathcal{B}}) \cdot u(1 + 2 + \cdots + (b/u - 1)) \cdot \bar{\chi} \\ &= \delta d \cdot \#\mathrm{Ker}(\psi_{\mathcal{B}}) \cdot b(b/u - 1)/2 \cdot \bar{\chi}.\end{aligned}$$

Denote

$$\begin{aligned}m &:= [K : K^g] = [K : K \cap (L.K^g)] \cdot [K \cap (L.K^g) : K^g] \\ &= b \cdot [K \cap (L.K^g) : K^g].\end{aligned}$$

Note that $[L.K^g : K^g] = [L : L_g] = d$, we denote

$$\begin{aligned}d' &:= d/[K \cap (L.K^g) : K^g] = [L.K^g : K \cap (L.K^g)] \\ &= [L.K : K] = [L : L \cap K].\end{aligned} \tag{3.5}$$

So we have

$$\partial_{A_g}(f(\mathcal{B})) = \delta d' \cdot \#\mathrm{Ker}(\psi_{\mathcal{B}}) \cdot m(b/u - 1)/2 \cdot \bar{\chi}. \tag{3.6}$$

Since $m \cdot \chi = 0$, we have

$$2\partial_{A_g}(f(\mathcal{B})) = 0,$$

this concludes the proof that $2\mathcal{B} \in \mathrm{Br}(X^c)$.

Our work shows that, if $2 \mid \delta d' \cdot \#\mathrm{Ker}(\psi_{\mathcal{B}}) \cdot (b/u - 1)$ in the formula (3.6), then $\partial_A(\mathcal{B}) = 0$ for any A, hence $\mathcal{B} \in \mathrm{Br}(X^c)$. We will prove the part (b) case by case.

(i) If $i = 0$, then the order $\mathrm{Gal}(K/k)$ is odd, hence b/u $(= [L_g : k])$ is odd. Therefore, $2 \mid (b/u - 1)$.

Suppose $i > 0$, write $\mathrm{Gal}(K/k) = H_1 \times H_2$, where H_1 is cyclic and of order 2^i and H_2 has odd order. By the Künneth formula [**31**, p. 96], we have

$$H^2(K/k, \mathbb{Q}/\mathbb{Z}) = \bigoplus_{i+j=2} H^i(H_1, H^j(H_2, \mathbb{Q}/\mathbb{Z})).$$

Since the orders of H_1 and H_2 are relatively prime, we have

$$H^1(H_1, H^1(H_2, \mathbb{Q}/\mathbb{Z})) = 0.$$

Obviously, $H^2(H_1, \mathbb{Q}/\mathbb{Z}) = 0$ since H_1 is cyclic. Then

$$H^2(K/k, \mathbb{Q}/\mathbb{Z}) = H^2(H_2, \mathbb{Q}/\mathbb{Z}).$$

We have

$$H^1(k, \mathrm{Pic}(\bar{X})) \simeq Ш^2_\omega(\hat{T})_P$$

by the sequence (2.2) and Proposition 2.2. Denote $h = \#H_2$, then

$$h \cdot H^1(k, \mathrm{Pic}(\bar{X})) = 0$$

by the fact $Ш^2_\omega(\hat{T}) \simeq H^3(K/k, \mathbb{Z}) \simeq H^2(K/k, \mathbb{Q}/\mathbb{Z})$.
Let $\mathcal{B} \in \mathrm{Br}(X)$. Since $\mathrm{Br}(X)/\mathrm{Br}_0(X) \hookrightarrow H^1(k, \mathrm{Pic}(\bar{X}))$ and $\partial_A(\mathrm{Br}_0(X)) = 0$, one has

$$h\partial_A(\mathcal{B}) = 0 \in H^1(\bar{\kappa}_A/\kappa_A, \mathbb{Q}/\mathbb{Z}).$$

On the other hand, we have

$$2\partial_A(\mathcal{B}) = 0 \in H^1(\bar{\kappa}_A/\kappa_A, \mathbb{Q}/\mathbb{Z})$$

by the proof of part (a). Since $(2, h) = 1$, we have $\partial_A(\mathcal{B}) = 0$, hence $\mathcal{B} \in \mathrm{Br}(X^c)$.

(ii) Since $L \cap K/k$ has odd degree, then L_g/k also has odd degree, that is, b/u is odd, then $2 \mid (b/u - 1)$.

(iii) Obviously, $2 \mid d'$ by (3.5).

(iv) If $2^s \nmid [L \cap K : k]$, then $[L : L \cap K]$ is even, hence $2 \mid d'$. So we may assume $2^s \mid [L \cap K : k]$.
Denote by $v_2(n)$ the 2-adic valuation for any $n \in \mathbb{Z}$, and denote $i := v_2([L \cap K : L_g])$. Obviously

$$v_2([L_g : k]) = v_2([L \cap K : k]) - v_2([L \cap K : L_g]) \geqslant s - i. \tag{3.7}$$

Since K/K^g is cyclic and the assumption for $\mathrm{Gal}(K/k)$, we have $v_2([K : K^g]) \leqslant s - 1$. Recall $b = [K : K \cap (L.K^g)]$, and note that

$$(L \cap K).K^g \subset K \cap (L.K^g) \text{ and } [(L \cap K).K^g : K^g] = [L \cap K : L_g],$$

we have

$$\begin{aligned} v_2(b) &\leqslant v_2([K : (L \cap K).K^g]) = v_2([K : K_g]) - v_2([(L \cap K).K^g : K^g]) \\ &\leqslant s - i - 1. \end{aligned} \tag{3.8}$$

By (3.7) and (3.8), the kernel of $\psi_{\mathcal{B}} : \mathrm{Gal}(L_g/k) \to \mathbb{Z}/b\mathbb{Z}$ has order divisible by 2, that is, $2 \mid \#\mathrm{Ker}(\psi_{\mathcal{B}})$.

(v) If $L' \not\subset L \cap K$, then $2 \mid [L : L \cap K]$, that is, $2 \mid d'$. So we may assume $L' \subset L \cap K$. Since K/K^g is cyclic, $L \cap K/L_g$ is also cyclic. Then there is a subfield L'' of $L_g \cap L'$ with $\mathrm{Gal}(L''/k) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, hence $2 \mid \#\mathrm{Ker}(\psi_{\mathcal{B}})$. □

Let X be the variety in the following proposition, then $\mathrm{Br}(X)/\mathrm{Br}_0(X) = \mathbb{Z}/n$. However, if n is even, $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c) = \mathbb{Z}/(n/2)$ by the following proposition, hence $\mathrm{Br}(X^c) \neq \mathrm{Br}(X)$ in this case.

PROPOSITION 3.7. *Suppose* $\mathrm{char}(k) = 0$ *and* $H^3(k, \bar{k}^\times) = 0$. *Let* n *be an integer. Let* K/k *be an abelian extension with* $\mathrm{Gal}(K/k) = \mathbb{Z}/n \times \mathbb{Z}/n$. *Suppose* $L \subset K$ *and* L/k *is cyclic of order* n. *Let* $P(t)$ *be an irreducible polynomial over* k *such that* $L \cong k[t]/(P(t))$. *Let* X *be the CTHS partial compactification for* $P(t)$ *and* K (*see Section* 1 *for definition*). *Then*

$$\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c) = \begin{cases} \mathbb{Z}/(n/2) & \text{if } n \text{ is even,} \\ \mathbb{Z}/n & \text{if } n \text{ is odd.} \end{cases}$$

Proof. Since K/L is cyclic, we have

$$Ш^2_\omega(\hat{T})_P = Ш^2_\omega(\hat{T}) \simeq H^3(\mathbb{Z}/n \times \mathbb{Z}/n, \mathbb{Z}) \simeq \mathbb{Z}/n,$$

the last equation follows from the Künneth formula [**31**, p. 96]. By the sequence (2.2) and Proposition 2.2, we have

$$H^1(k, \mathrm{Pic}(\bar{X})) \simeq Ш^2_\omega(\hat{T})_P \simeq \mathbb{Z}/n.$$

Since $H^3(k, \bar{k}^\times) = 0$, we have

$$\mathrm{Br}(X)/\mathrm{Br}_0(X) \simeq H^1(k, \mathrm{Pic}(\bar{X})) \simeq \mathbb{Z}/n.$$

If n is odd, then the result immediately follows from the case (a) in Theorem 3.6. Then we only need to consider the case n is even.

Write $n = 2^s n_0$, where n_0 is odd and $s \geqslant 1$. By Theorem 3.6, the cokernel of $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$ in $\mathrm{Br}(X)/\mathrm{Br}_0(X)$ is killed by 2. Then we only need to show that an element B in $\mathrm{Br}(X)/\mathrm{Br}_0(X)$ with order 2^s is not contained in $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$.

Let $\langle g \rangle$ be the cyclic subgroup of $\mathrm{Gal}(K/k)$ generated by g. Let K^g be the fixed field of $\langle g \rangle$ in K. We have the following morphism:

$$f_g : H^1(K/k, \mathrm{Pic}(X_K)) \longrightarrow H^1(K/K^g, \mathrm{Pic}(X_K)).$$

Let T' be the k-torus defined by $N_{L/k}(\Xi_1) \cdot N_{K/k}(\Xi_2) = 1$. Let $f := \prod_{\langle g \rangle} f_g$. Since

$$\mathrm{Ker}(f) = Ш^1_\omega(K/k, \mathrm{Pic}(X_K)) = Ш^1_\omega(\mathrm{Pic}(\bar{X})) = Ш^2_\omega(\hat{T}') = 0$$

by Proposition 2.3 for the last second equality and Corollary 3.3 for the last equality, $f(B)$ has order 2^s.

There exists $\langle g_0 \rangle$ such that $f_{g_0}(B)$ has order divisible by 2^s since $f(B)$ has order 2^s. Denote $L_{g_0} := L \cap K^{g_0}$. Then both $K/L.K^{g_0}$ and L_{g_0}/k have order divisible by 2^s by Lemma 3.5, since $f_{g_0}(B)$ has order divisible by 2^s. Let Θ_0/k be the unique subfield of K with $\mathrm{Gal}(\Theta_0/k) = \mathbb{Z}/n_0 \times \mathbb{Z}/n_0$ and let $\Theta = K^{g_0}.\Theta_0$. Then we can see $\Theta = L.\Theta_0$ and K/Θ has degree 2^s.

By the inflation–restriction sequence, the kernel of the map

$$H^1(K/K^{g_0}, \mathrm{Pic}(X_K)) \longrightarrow H^1(K/\Theta, \mathrm{Pic}(X_K))$$

has odd order since Θ/K^{g_0} has odd degree. Since $f_{g_0}(B) \in H^1(K/K^{g_0}, \mathrm{Pic}(X_K))$ has order divisible by 2^s and K/Θ has degree 2^s, $f_\Theta(B) \in H^1(K/\Theta, \mathrm{Pic}(X_K))$ has order 2^s, where $f_\Theta : H^1(K/k, \mathrm{Pic}(X_K)) \to H^1(K/\Theta, \mathrm{Pic}(X_K))$.

Let $\mathcal{B} \in \mathrm{Br}(X)$ which lifts B. Let $f'_\Theta : \mathrm{Br}(X) \to \mathrm{Br}(X_\Theta)$ be the natural map. By Lemma 3.5, we have

$$f'_\Theta(\mathcal{B}) = \rho + \sum_{\sigma \in \mathrm{Gal}(L/k)} \psi_{\mathcal{B}}(\sigma)(t - \sigma(\eta), \chi),$$

where $\rho \in \mathrm{Br}(\Theta)$, η is a root of $P(t)$ in L, $\psi_{\mathcal{B}} \in \mathrm{Hom}(\mathrm{Gal}(L/k), \mathbb{Z}/2^s)$ and $\chi \in \mathrm{Hom}(\mathrm{Gal}(K/\Theta), \mathbb{Z}/2^s)$ is primitive. Since $f_\Theta(B)$ has order 2^s, we deduce $\psi_{\mathcal{B}}$ has order divisible by 2^s. Since $\mathrm{Gal}(L/k)$ is cyclic of order $n = 2^s \cdot n_0$ with n_0 odd, then $\psi_{\mathcal{B}}$ has order 2^s.

The variety X_Θ contains an open affine Θ-subvariety U defined by

$$\prod_i N_{K/\Theta}(\Xi_i) = P(t) \quad \text{and } t \neq 0,$$

where $K \otimes_k \Theta \cong \prod_i K$ and K/Θ has degree 2^s. Let $Q(u) = u^n P(1/u)$. Let W be the smooth affine Θ-variety defined by

$$\prod_i N_{K/\Theta}(\Xi'_i) = Q(u).$$

Obviously, the open subvariety of W defined by $u \neq 0$ is isomorphic to U by the map $u \mapsto 1/t, \Xi'_1 \mapsto \Xi_1/t^\mu, \Xi'_i \mapsto \Xi_i$ for $i \geqslant 2$, where $\mu = n/2^s$.

Let D be the divisor of W over Θ defined by $u = 0$. It is easy to see that the divisor D is geometrically irreducible. So we have $\bar{\Theta} \cap \kappa_D = \Theta$, where κ_D is the function field of D. The local ring A_D associated with D is a discrete valuation ring with $\mathrm{ord}_{A_D}(t) = -1$ and $\kappa_D = \kappa_{A_D}$. So $\mathrm{ord}_{A_D}(t - \sigma(\eta)) = -1$ for all σ. Therefore,

$$\partial_{A_D}(f'_\Theta(\mathcal{B})) = -\sum_{\sigma \in \mathrm{Gal}(L/k)} \psi_{\mathcal{B}}(\sigma) \cdot \bar{\chi}.$$

Since $\psi_{\mathcal{B}}$ has order 2^s, then $\psi_{\mathcal{B}} : \mathrm{Gal}(L/k) \to \mathbb{Z}/2^s$ is surjective, hence

$$\sum_{\sigma \in \mathrm{Gal}(L/k)} \psi_{\mathcal{B}}(\sigma) = n_0 \cdot 2^{s-1}(2^s - 1) \equiv -n_0 \cdot 2^{s-1} \in \mathbb{Z}/2^s.$$

Since χ has order 2^s, we have

$$\sum_{\sigma \in \mathrm{Gal}(L/k)} \psi_{\mathcal{B}}(\sigma) \cdot \chi = -n_0 \cdot 2^{s-1} \cdot \chi \neq 0 \in \mathrm{Hom}(\mathrm{Gal}(K/\Theta), \mathbb{Q}/\mathbb{Z}).$$

Since $\bar{\Theta} \cap \kappa_{A_D} = \Theta$, we have

$$\partial_{A_D}(f'_\Theta(\mathcal{B})) \neq 0 \in H^1(\kappa_{A_D}, \mathbb{Q}/\mathbb{Z}),$$

Then $f'_\Theta(\mathcal{B}) \notin \mathrm{Br}(X^c_\Theta)$, hence $\mathcal{B} \notin \mathrm{Br}(X^c)$. □

REMARK 3.8. In the case $n = 2$, we obtain $\mathrm{Br}(X^c) = \mathrm{Br}_0(X^c)$, this answers the final question in [8, Questions on p. 82 and p. 83].

3.3. *A case with K/k a Galois extension and $P(t)$ with multiple rational roots*

The part of this section is to prove Proposition 3.10, which is motivated by the Question (a) in [8, p. 82].

Let G be a finite group and M a G-module. Define

$$Ш^2_\omega(G, M) := \bigcap_{g \in G} \mathrm{Ker}[H^2(G, M) \to H^2(\langle g \rangle, M)].$$

LEMMA 3.9. *Let p be a prime, $G = \mathbb{Z}/p \times \mathbb{Z}/p$ and $\mathbb{Z}/p$ the trivial G-module. Then*

$$Ш^2_\omega(G, \mathbb{Z}/p) = \begin{cases} 0 & \text{if } p = 2, \\ \mathbb{Z}/p & \text{if } p \text{ is odd.} \end{cases}$$

Proof. Denote $R := \mathbb{Z}/p$. Let $H_1 = \langle g_1 \rangle, H_2 = \langle g_2 \rangle$ and $G = H_1 \times H_2$. Let $N = N_1 + N_2$ with $N_i = \sum_{\sigma \in G/H_i} \sigma$ for $i = 1, 2$. Denote

$$\Delta := (R[G/H_1] \oplus R[G/H_2])/(R \cdot N).$$

Then we have the following exact sequence:

$$0 \longrightarrow R \longrightarrow R[G/H_1] \oplus R[G/H_2] \longrightarrow \Delta \longrightarrow 0.$$

Since the restriction map

$$H^1(G, \mathbb{Z}/p) \longrightarrow H^1(H_1, \mathbb{Z}/p) \times H^1(H_2, \mathbb{Z}/p) \quad \text{is surjective,}$$

we have the following exact sequence:

$$0 \longrightarrow H^1(G, \Delta) \longrightarrow H^2(G, \mathbb{Z}/p) \longrightarrow H^2(H_1, \mathbb{Z}/p) \times H^2(H_2, \mathbb{Z}/p).$$

Let H be a non-trivial cyclic subgroup of G and $H \neq H_1, H_2$. It is easy to verify that $R[G/H_i] \simeq R[H]$ as H-modules for $i = 1, 2$. Then

$$H^i(H, R[G/H_1] \times R[G/H_2]) = 0 \quad \text{if } i \geqslant 1.$$

Therefore, we have the following commutative diagram:

$$\begin{array}{ccccccc} 0 & \twoheadrightarrow & H^1(G, \Delta) & \twoheadrightarrow & H^2(G, \mathbb{Z}/p) & \twoheadrightarrow & H^2(H_1, \mathbb{Z}/p) \oplus H^2(H_2, \mathbb{Z}/p) \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \twoheadrightarrow & H^1(H, \Delta) & \twoheadrightarrow & H^2(H, \mathbb{Z}/p) & \longrightarrow & 0 \end{array}$$

Therefore, we have

$$Ш^2_\omega(G, \mathbb{Z}/p) \cong \bigcap_{H \neq H_1, H_2} \mathrm{Ker}[H^1(G, \Delta) \to H^1(H, \Delta)].$$

By the inflation–restriction sequence, we have

$$H^1(G/H, \Delta^H) = \mathrm{Ker}[H^1(G, \Delta) \to H^1(H, \Delta)].$$

Since G/H is cyclic and Δ is finite, using Tate cohomology one has

$$| H^1(G/H, \Delta^H) | = | \hat{H}^0(G/H, \Delta^H) | = | \Delta^G / N_{G/H}(\Delta^H) | .$$

It is easy to verify that $\Delta^G \simeq \mathbb{Z}/p$ by the exact sequence

$$0 \longrightarrow \mathbb{Z}/p \longrightarrow R[G/H_1] \oplus R[G/H_2] \longrightarrow \Delta \longrightarrow 0.$$

Furthermore, we have

$$0 \longrightarrow \mathbb{Z}/p \longrightarrow \mathbb{Z}/p \cdot N_1 \oplus \mathbb{Z}/p \cdot N_2 \longrightarrow \Delta^H \longrightarrow H^1(H, \mathbb{Z}/p) \longrightarrow 0.$$

By a similar argument as for equation (3.2), Δ^H is generated by

$$R \cdot N_1 \oplus R \cdot N_2 \text{ and } \left\{ \sum_{g \in H} \chi(g)(\bar{g}, \bar{g}) : \chi \in H^1(H, \mathbb{Z}/p) \right\},$$

where $(\bar{g}, \bar{g}) \in G/H_1 \times G/H_2$. Since G/H trivially acts on $R \cdot N_1 \oplus R \cdot N_2$, we have

$$N_{G/H}(R \cdot N_1 \oplus R \cdot N_2) = p \cdot (R \cdot N_1 \oplus R \cdot N_2) = 0.$$

Note that $H_1 \simeq G/H$ since $G = \mathbb{Z}/p \times \mathbb{Z}/p$ and p is a prime. Denote

$$u = \sum_{g \in H} \chi(g)(\bar{g}, \bar{g}) \in \Delta, \text{ where } \chi \text{ is a primitive character.}$$

Then

$$\sigma(u) = \sum_{g \in H} \chi(g)(\bar{g}, \overline{\sigma g}), \quad \text{where } \sigma \in H_1.$$

Hence we have

$$N_{G/H}(u) = \sum_{\sigma\in H_1}\sum_{g\in H}\chi(g)(\bar{g},\overline{\sigma g}) = \sum_{g\in H}\chi(g)\sum_{\sigma\in H_1}(\bar{g},\overline{\sigma g})$$
$$= \sum_{g\in H}\chi(g)\left(0,\sum_{\sigma\in H_1}\overline{\sigma g}\right) = \left(0,\sum_{g\in H}\chi(g)N_2\right)$$
$$= \begin{cases}(0,N_2) & \text{if } p=2,\\ 0 & \text{if } p \text{ is odd.}\end{cases}$$

Therefore, we have

$$\mid H^1(G/H,\Delta^H)\mid = \mid \Delta^G/N_{G/H}(\Delta^H)\mid = \begin{cases}0 & \text{if } p=2,\\ p & \text{if } p \text{ is odd.}\end{cases}$$

If $p=2$, then we have $Ш^2_\omega(G,\mathbb{Z}/p)=0$.

By the Künneth formula [**31**, p. 96], we have

$$H^2(G,\mathbb{Z}/p)\simeq\bigoplus_{i+j=2}H^i(H_1,H^j(H_2,\mathbb{Z}/p)).$$

Therefore,

$$H^1(G,\Delta)\simeq \mathrm{Ker}[H^2(G,\mathbb{Z}/p)\to H^2(H_1,\mathbb{Z}/p)\oplus H^2(H_2,\mathbb{Z}/p)]\simeq\mathbb{Z}/p.$$

Suppose that p is odd. By the above argument,

$$\mathrm{Ker}[H^1(G,\Delta)\to H^1(H,\Delta)]\simeq H^1(G/H,\Delta^H)\simeq\mathbb{Z}/p.$$

Therefore, the restriction map $H^1(G,\Delta)\to H^1(H,\Delta)$ is a zero map, hence $Ш^2_\omega(G,\mathbb{Z}/p)\simeq H^1(G,\Delta)\simeq\mathbb{Z}/p$. □

PROPOSITION 3.10. *Suppose $H^3(k,\bar{k}^\times)=0$. Let $P(t)=c\prod_{i=1}^m(t-e_i)^{d_i}$ with $c\in k^\times$, $e_i\in k$ distinct and $d:=gcd(d_1,\dots,d_m)$. Let K/k be a Galois extension with $\mathrm{Gal}(K/k)$ killed by d. Let X be the CTHS partial compactification for $P(t)$ and K (see Section 1 for definition), and X^c a smooth compactification of X. Then*

$$\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)\cong Ш^2_\omega(K/k,\mathbb{Z}/d).$$

Proof. Let $\langle g\rangle\subset\mathrm{Gal}(K/k)$ be a cyclic subgroup. Let K^g be the fixed field of $\langle g\rangle$ in K. We have the following morphism:

$$f_g:\mathrm{Br}(X)/\mathrm{Br}_0(X)\longrightarrow\mathrm{Br}(X_{K^g})/\mathrm{Br}_0(X_{K^g}).$$

Since $[K:K^g]\mid d$ by our assumption, the K^g-variety X_{K^g} is K^g-birationally isomorphic to $\mathbb{A}^1_{K^g}\times Y$, where Y is a K^g-variety defined by the equation $c=\prod_i N_{K/K^g}(\Xi_i)$ and $K\otimes_k K^g\cong\prod_i K$. We know $H^1(K^g,\mathrm{Pic}(\bar{Y}^c))=0$ since K/K^g is cyclic, which implies $\mathrm{Br}(X^c_{K^g})/\mathrm{Br}_0(X^c_{K^g})=0$.

If $B\in\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$, then we have $f_g(B)\in\mathrm{Br}(X^c_{K^g})/\mathrm{Br}_0(X_{K^g})$, hence $f_g(B)=0\in\mathrm{Br}(X_{K^g})/\mathrm{Br}_0(X^c_{K^g})$. Let $f:=\prod_{\langle g\rangle}f_g$, so

$$\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)\subset\mathrm{Ker}(f).$$

Let T' be the torus over k defined by $t_1^{d_1}\cdots t_m^{d_m}N_{K/k}(\Xi)=1$. By Proposition 2.3, we know $\mathrm{Ker}(f)\simeq Ш^2_\omega(K/k,\hat{T}')$ is contained in $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$ since $H^3(k,\bar{k}^\times)=0$. So $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)=\mathrm{Ker}(f)$ and we only need to show

$$Ш^2_\omega(K/k,\hat{T}')\simeq Ш^2_\omega(K/k,\mathbb{Z}/d).$$

Let $N = \sum_{\sigma\in \mathrm{Gal}(K/k)} \sigma$. We know

$$\hat{T}' \simeq [(\mathbb{Z}\oplus\cdots\oplus\mathbb{Z})\oplus\mathbb{Z}[K/k]]/\mathbb{Z}\cdot((d_1,\ldots,d_m)+N).$$

Denote

$$M := (\mathbb{Z}\oplus\cdots\oplus\mathbb{Z})/\mathbb{Z}\cdot(d_1,\ldots,d_m).$$

This exact sequence

$$0\longrightarrow\mathbb{Z}[K/k]\longrightarrow\hat{T}'\longrightarrow M\longrightarrow 0$$

implies $Ш^2_\omega(K/k,\hat{T}')\simeq Ш^2_\omega(K/k,M)$. Since $M\cong\mathbb{Z}/d\times M'$, where M' is a free $\mathbb{Z}$-module, we have $Ш^2_\omega(K/k,M)\simeq Ш^2_\omega(K/k,\mathbb{Z}/d)$ by the fact $Ш^2_\omega(K/k,M')=0$. □

By Lemma 3.9 and Proposition 3.10, we have the following result.

Corollary 3.11. *Suppose $H^3(k,\bar{k}^\times)=0$. Let p be a prime and $P(t)=c\prod_{i=1}^m(t-e_i)^p$ with $c\in k^\times$, $e_i\in k$ distinct. Let K/k be an abelian extension with $\mathrm{Gal}(K/k)=\mathbb{Z}/p\times\mathbb{Z}/p$. Then*

$$\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)=\begin{cases}0 & \text{if } p=2,\\ \mathbb{Z}/p & \text{if } p \text{ is odd.}\end{cases}$$

Remark 3.12. For $p=2$, this corollary answers Question (a) in [8, p. 82].

4. *Rational points under Schinzel's hypothesis*

Let k be a number field. Let Ω_k be the set of all places of k. Let $P(t)$ be a non-zero polynomial over k. In this section, we mainly consider the question whether the Brauer–Manin obstruction is the only obstruction to the Hasse principle and weak approximation for a smooth and proper model of the k-variety defined by

$$N_{K/k}(\Xi)=P(t),$$

where Ξ is a 'variable' in K, K/k is a finite field extension and $N_{K/k}$ is a formal norm associated with the extension for the variable Ξ.

A positive answer was given by Colliot-Thélène, Sansuc and Swinnerton-Dyer in their remarkable paper [16] when $P(t)$ has degree $\leqslant 4$ and K/k is a quadratic extension. A positive answer is also known for the following cases: the extension K/k has degree 3 and the polynomial $P(t)$ has degree at most 3 [10]; the polynomial $P(t)$ having just two roots in k and K/k arbitrary [8, 23, 39]; $k=\mathbb{Q}$, $P(t)$ has degree 2 and $K/\mathbb{Q}$ arbitrary [1, 20]; $k=\mathbb{Q}$, $P(t)$ is split in $\mathbb{Q}$ and $K/\mathbb{Q}$ arbitrary [2, 3, 22].

Conditional results have been obtained under Schinzel's hypothesis (H). Under this hypothesis, Colliot-Thélène, Skorobogatov and Swinnerton-Dyer [17, Theorem 1.1 (e)] proved that the Brauer–Manin obstruction is the only obstruction to the Hasse principle and weak approximation for smooth projective models of varieties defined by an equation $N_{K/k}(\Xi)=P(t)$ when the extension K/k is abelian and norm equations $N_{K/k}(\Xi)=c$ (for any $c\in k^\times$) satisfy the Hasse principle and weak approximation, for instance when K/k is cyclic. Their result is more general, they consider smooth projective varieties over k which admit a fibration over the projective line such that:

(a) a certain abelianity condition on the splitting field of the singular fibres holds (Condition (i) in [17, Theorem 1.1]);
(b) the Hasse principle and weak approximation hold on the smooth fibres.

In this section, we shall handle three new classes of fibrations over the projective line whose generic fibre is birationally a principal homogeneous space under a torus. In each of these classes one of conditions (a) or (b) is not in general fulfilled.

The following observation is important for the proof of Theorems 4.2 and 5.1.

LEMMA 4.1. *Let k be a number field and K/k a finite field extension. Let $P(t)$ be a non-zero polynomial over k. Let X be the CTHS partial compactification (see Section 1 for definition) of the equation $N_{K/k}(\Xi) = P(t)$ and $X_K := X \times_k K$. Then $\mathrm{Br}(X_K) = \mathrm{Br}_0(X_K)$.*

Proof. Denote by T the torus $R^1_{K/k}(\mathbb{G}_m)$. Let $T_K := T \times_k K$, then $\hat{T}_K$ is a permutation Γ_K-module, hence $Ш^2_\omega(K, \hat{T}_K) = 0$ and $H^1(K, \mathbb{Z}_P \otimes \hat{T}_K) = 0$. Therefore, $H^1(K, \mathrm{Pic}(\bar{X})) = 0$ by the sequence (2.2), hence $\mathrm{Br}(X_K) = \mathrm{Br}_0(X_K)$. □

THEOREM 4.2. *Let k be a number field and K/k an abelian extension. Let $P(t)$ be a non-zero polynomial over k and $T = R^1_{K/k}(\mathbb{G}_m)$. Suppose $Ш^2_\omega(\hat{T})_P = Ш^2_\omega(\hat{T})$ (see Section 1 for definition). Assume Schinzel's hypothesis holds. Then the Brauer–Manin obstruction to the Hasse principle and weak approximation for rational points is the only obstruction for any smooth proper model of the variety over k defined by the equation*

$$N_{K/k}(\Xi) = P(t). \tag{4.1}$$

Proof. It is sufficient to prove the statement for any given model. Let V be the smooth locus of the affine k-variety defined by (4.1). Let Y be a smooth compactification of V with a projection $p : Y \to \mathbb{P}^1_k$ defined by $(\Xi, t) \mapsto t$.

We assume that Y has points in all completions of k, and we are given a finite set S of places of k which contains all archimedean places, and points $P_v \in Y(k_v)$ for $v \in S$. We assume that there is no Brauer–Manin obstruction to weak approximation for $(P_v)_{v\in S}$. This means that we may complete the family $(P_v)_{v\in S}$ to a family $(P_v)_{v\in\Omega_k}$ such that

$$\text{for any } \mathcal{B} \in \mathrm{Br}(Y), \sum_{v\in\Omega_k} \mathrm{inv}_v(\mathcal{B}(P_v)) = 0 \in \mathbb{Q}/\mathbb{Z}. \tag{4.2}$$

From this, we want to deduce that there exists $P \in Y(k)$ as close as we wish to each $P_v \in Y(k_v)$ for $v \in S$.

Let $U_0 \subset \mathbb{A}^1_k$ be a non-empty Zariski open set defined by $P(t) \neq 0$. By the continuity of the pairing of $Y(k_v)$ with elements of the Brauer group, we can replace each P_v by a close enough point such that condition (4.2), which only involves finitely many classes in the Brauer group, will not be affected. Therefore, we can assume that all points appearing in (4.2) lie in $p^{-1}(U_0)$ by the implicit function theorem.

Let $Q_v := p(P_v) \in U_0(k_v) \subset \mathbb{A}^1(k_v)$. We first choose a k-point $Q_0 \in U_0(k)$, close enough to each Q_v for each archimedean place v, and such that Q_0 is different from each Q_v for $v \in S$. Let $u = 1/(t - Q_0)$. Denote $Q(u) := u^{\deg(P(t))}P(1/u + Q_0)$ and $n := [K : k]$. Choose l sufficient large such that $\ln - \deg(P(t)) > 0$ and denote $P^{(1)}(t) := t^{\ln - \deg(P(t))}Q(t)$. Then we obtain a new smooth affine variety W defined by

$$N_{K/k}(\Xi) = P^{(1)}(t) \neq 0,$$

and it is isomorphic to the open subvariety of $p^{-1}(U_0)$ defined by $t \neq Q_0$ which contains all P_v for $v \in S$. Let $W_0 \subset \mathbb{A}^1_k$ be a non-empty Zariski open subset defined by $P^{(1)}(t) \neq 0$. Then p projects W to W_0.

We are now looking for a point $Q \in W_0(k)$ with associated coordinate $\lambda \in k$, such that λ is very close to each Q_v for $v \in S$, v finite, λ is big enough at each archimedean place of k (Q is close enough to the infinite point), and such that the fibre W_Q of p has a k-rational point.

Let X be the *CTHS* partial compactification (see Section 1 for definition) of V. We know $H^1(k, \mathrm{Pic}(\bar{X})) \to Ш^2_\omega(\hat{T})$ is surjective by our assumption and the sequence (2.2). Then we can choose a finite subset $B \subset \mathrm{Br}(X)$ such that the image of B by the composite map

$$\mathrm{Br}(X) \longrightarrow H^1(k, \mathrm{Pic}(\bar{X})) \longrightarrow Ш^2_\omega(\hat{T})$$

is $Ш^2_\omega(\hat{T})$.

Choose $\omega_1, \ldots, \omega_n \in \mathfrak{o}_K$ to be a basis of K over k, where $\mathfrak{o}_K$ (respectively, $\mathfrak{o}_k$) is the ring of integers of K (respectively, k). Choose S' to be a finite set of places of k containing all archimedean places such that $P^{(1)}(t) \in \mathfrak{o}_{S'}[t]$ and $\omega_1, \ldots, \omega_n$ generate $\mathfrak{o}_K \otimes_{\mathfrak{o}_k} \mathfrak{o}_{k_v}$ over $\mathfrak{o}_{k_v}$ for any $v \notin S'$. Let $\mathcal{W} \subset \mathbb{A}^{n+2}_{\mathfrak{o}_{S'}}$ be the smooth affine integral model of W over $\mathfrak{o}_{S'}$ defined by

$$N_{K/k}(x_1\omega_1 + \cdots + x_n\omega_n) = P^{(1)}(t) \quad \text{and } y \cdot P^{(1)}(t) = 1.$$

We enlarge S' such that for any $\mathcal{B} \in B$, $\mathcal{B}$ can be extended to $\mathrm{Br}(\mathcal{W})$.

Now we enlarge S such that $S \supset S'$, and that S contains all the original places at which we want to approximate, and that S also contains the places associated in Hypothesis H_1 (see [**18**, p. 71]) to the polynomials of $P^{(1)}(t)$, all ramified places of K/k and all places v such that for some $\mathcal{B} \in B$, $\mathrm{inv}_w(j_K(\mathcal{B})) \neq 0$, where w is a place of K over v and $j_K : \mathrm{Br}(X) \to \mathrm{Br}(X_K)$, noting that $\mathrm{Br}(X_K) = \mathrm{Br}_0(X_K)$ by Lemma 4.1.

Let $\{p_i(t) \mid 1 \leqslant i \leqslant m\}$ be all irreducible terms of $P^{(1)}(t)$ over k and let

$$A = \{(p_i(t), \chi) \in \mathrm{Br}(W) : 1 \leqslant i \leqslant m, \chi \in \mathrm{Hom}(\mathrm{Gal}(K/k), \mathbb{Q}/\mathbb{Z})\}.$$

Since condition (4.2) holds, there is no Brauer–Manin obstruction on Y to weak approximation for $(P_v)_{v \in S}$. According to Harari's formal lemma (see [**21**]), we may find a finite set S_1 of places of k, containing S, and points $P_v \in W(k_v)$, $v \in S_1$, and which extend the given family

$$P_v \in W(k_v), v \in S,$$

such that for each $\mathcal{B} \in A \cup B$

$$\sum_{v \in S_1} \mathrm{inv}_v(\mathcal{B}(P_v)) = 0. \tag{4.3}$$

Now apply Hypothesis (H_1) (see [**18**, Proposition 4.1]), we thus find $\lambda \in k$ close enough to each Q_v for all finite places $v \in S_1$, λ integral away from S_1, and λ as large as one wants at all archimedean places, so that:

(i) The fibre W_λ of p contains a k_v-point P'_v which is as close as we wish to P_v for all places $v \in S_1$, and such that

$$\mathrm{inv}_v(\mathcal{B}(P'_v)) = \mathrm{inv}_v(\mathcal{B}(P_v))$$

for each $\mathcal{B} \in A \cup B$ and $v \in S_1$.

(ii) For each irreducible term $p_i(t)$ of the polynomial $P^{(1)}(t)$, there exists a place v_i such that $p_i(\lambda)$ is a uniformizer of $\mathfrak{o}_{k_{v_i}}$ and is a unit in $\mathfrak{o}_{k_v}$ if $v \notin S_1$ and $v \neq v_i$.

For each $\mathcal{B} \in A \cup B$, by (4.3) and point (i) we have

$$0 = \sum_{v \in S_1} \mathrm{inv}_v(\mathcal{B}(P_v)) = \sum_{v \in S_1} \mathrm{inv}_v(\mathcal{B}(P'_v)). \tag{4.4}$$

In particular, for each $\mathcal{B} = (p_i(\lambda), \chi) \in A$, we have

$$\sum_{v \in S_1} \mathrm{inv}_v(p_i(\lambda), \chi) = 0.$$

Since the sum of all local invariants inv_v of $(p_i(\lambda), \chi)$ over k vanishes (global class field theory), we deduce

$$\sum_{v \notin S_1} \mathrm{inv}_v(p_i(\lambda), \chi) = 0. \tag{4.5}$$

We have $\mathrm{inv}_v(p_i(\lambda), \chi) = 0$ for $v \notin S_1$ and $v \neq v_i$, since $p_i(\lambda)$ is a unit at v by point (ii). Then (4.5) implies

$$\mathrm{inv}_{v_i}(p_i(\lambda), \chi) = 0. \tag{4.6}$$

Since $p_i(\lambda)$ is a uniformizer at v_i and χ runs through all characters of $\mathrm{Gal}(K/k)$ and K/k is abelian, we know K/k is totally split at v_i by (4.6). Therefore, the fibre W_λ contains a k_{v_i}-point P'_{v_i} for all places v_i. Obviously, the fibre $\mathcal{W}_\lambda$ contains an $\mathfrak{o}_{k_v}$-point P'_v for all places $v \notin S_1$ and $v \neq v_i$ for any i, since all $p_i(\lambda)$ are units. So the fibre W_λ contains such a k_v-point P'_v for all places v of k.

In the following, we will show $\mathrm{inv}_v(\mathcal{B}(P'_v)) = 0$ for $v \notin S_1$ and $\mathcal{B} \in B$, hence

$$\sum_{v \in \Omega_k} \mathrm{inv}_v(\mathcal{B}(P'_v)) = \sum_{v \in S_1} \mathrm{inv}_v(\mathcal{B}(P'_v)) = 0 \tag{4.7}$$

by (4.4). Let $v \notin S_1$ and $v \neq v_i$ for $1 \leqslant i \leqslant m$. We have $\mathrm{inv}_v(\mathcal{B}(P'_v)) = 0$ since $\mathcal{B}$ can be extended to $\mathcal{W}$ and $P'_v \in \mathcal{W}(\mathfrak{o}_{k_v})$. If v is some v_i, then K/k is totally split at v by the above arguments. Since $\mathrm{Br}(X_K) = \mathrm{Br}_0(X_K)$ by Lemma 4.1, then the image of $\mathcal{B}$ in $\mathrm{Br}(X_{k_v})$ is trivial by the choice of S ($S_1 \supset S$), hence $\mathrm{inv}_v(\mathcal{B}(P'_v)) = 0$.

In the following, we will show the image of B by the induced map $\mathrm{Br}(X)/\mathrm{Br}(k) \to \mathrm{Br}(X_\lambda)/\mathrm{Br}(k)$ is surjective, then (4.7) implies there is no Brauer–Manin obstruction on W_λ for $(P'_v)_{v \in \Omega_k}$, hence we can find a k-point on W_λ to approximate $(P'_v)_{v \in S}$ by the property of the principal homogeneous space of tori [**35**, Theorem 8.12] or [**37**, Theorem 5.2.1].

The map $H^1(k, \mathrm{Pic}(\bar{X})) \to Ш^2_\omega(\hat{T})_P$ in the sequence (2.2) is induced by the map $\mathrm{Pic}(\bar{X}) \to \mathrm{Pic}(X_{\bar{\eta}})$, where η is the generic point of $\mathbb{P}^1_k$. We have the following commutative diagram:

$$\begin{array}{ccc} \mathrm{Pic}(X_{\bar{k}[t]_{(t-\lambda)}}) & \longrightarrow & \mathrm{Pic}(X_{\bar{k}(t)}) \\ \| & & \downarrow i \\ \mathrm{Pic}(X_{\bar{k}[t]_{(t-\lambda)}}) & \longrightarrow & \mathrm{Pic}(\bar{X}_\lambda) \end{array}$$

Therefore, we have the following commutative diagram:

$$\begin{array}{ccc} \mathrm{Pic}(\bar{X}) & \longrightarrow & \mathrm{Pic}(X_{\bar{k}(t)}) \\ \| & & \downarrow i \\ \mathrm{Pic}(\bar{X}) & \longrightarrow & \mathrm{Pic}(\bar{X}_\lambda) \end{array}$$

By [**8**, Lemma 2.1], we know the morphism $i : \mathrm{Pic}(X_{\bar{k}(t)}) \to \mathrm{Pic}(\bar{X}_\lambda)$ is an isomorphism as Γ_k-module. Since $\mathrm{Pic}(X_{\bar{\eta}})$ is torsion-free, we have the isomorphism

$$H^1(k(t), \mathrm{Pic}(X_{\bar{\eta}})) \cong H^1(k, \mathrm{Pic}(X_{\bar{k}(t)})) \cong H^1(k, \mathrm{Pic}(\bar{X}_\lambda)).$$

Therefore, we have the following commutative diagram:

$$\begin{array}{ccc} H^1(k, \mathrm{Pic}(\bar{X})) & \longrightarrow & H^1(k(t), \mathrm{Pic}(X_{\bar{\eta}})) \\ \| & & \downarrow \cong \\ H^1(k, \mathrm{Pic}(\bar{X})) & \longrightarrow & H^1(k, \mathrm{Pic}(\bar{X}_\lambda)) \end{array}$$

hence the image of B by the induced map $\mathrm{Br}(X)/\mathrm{Br}(k) \to \mathrm{Br}(X_\lambda)/\mathrm{Br}(k)$ is $\mathrm{Br}(X_\lambda)/\mathrm{Br}(k)$. □

REMARK 4.3. (i) The condition $Ш^2_\omega(\hat{T})_P = Ш^2_\omega(\hat{T})$ is equivalent to the condition that the natural morphism

$$\mathrm{Br}(X)/\mathrm{Br}(k) \longrightarrow \mathrm{Br}(X_\eta)/\mathrm{Br}(k(\eta))$$

is surjective, where X is the CTHS partial compactification with the projection $p : X \to \mathbb{A}^1$, and η is the generic point of $\mathbb{A}^1$.

(ii) Many similar results to Theorem 4.2 have been proved for more general varieties which have a fibration to $\mathbb{P}^1$, see [4, 27, 38].

As a direct application of this theorem, we have the following corollary.

COROLLARY 4.4. *Let n be a positive integer and let K/k be an abelian extension with $\mathrm{Gal}(K/k) = \mathbb{Z}/n \times \mathbb{Z}/n$. Let $P(t)$ be an irreducible polynomial over k and $L = k[t]/(P(t))$. Assume that L contains a cyclic subfield of K with degree n. Let V be the smooth affine variety over k defined by*

$$N_{K/k}(\Xi) = P(t).$$

Assume Schinzel's hypothesis holds, then the Brauer–Manin obstruction to the Hasse principle and weak approximation is the only obstruction for any smooth proper model of V.

Proof. Let $L = k[t]/(P(t))$ and T the torus $R^1_{K/k}(\mathbb{G}_m)$. By the exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}[K/k] \longrightarrow \hat{T} \longrightarrow 0,$$

we have

$$Ш^2_\omega(\hat{T}) = H^3(K/k, \mathbb{Z}) \quad \text{and} \quad Ш^2(\hat{T})_P = \mathrm{Ker}[H^3(K/k, \mathbb{Z}) \to H^3(K/L \cap K, \mathbb{Z})].$$

Since $K/L \cap K$ is cycic, we have

$$H^3(K/L \cap K, \mathbb{Z}) = H^1(K/L \cap K, \mathbb{Z}) = 0,$$

hence $Ш^2(\hat{T})_P = Ш^2(\hat{T})$. □

In the following theorem, the condition $Ш^2_\omega(\hat{T})_P = Ш^2_\omega(\hat{T})$ is not in general fulfilled.

THEOREM 4.5. *Let k be a number field and $P(t)$ a non-zero polynomial over k. Assume Schinzel's hypothesis holds. Then the Brauer–Manin obstruction to the Hasse principle and weak approximation for rational points is the only obstruction for any smooth proper model of the variety over k defined by the equation*

$$(x_1^2 - ax_2^2)(y_1^2 - by_2^2)(z_1^2 - abz_2^2) = P(t),$$

where $a, b \in k^\times$.

Proof. It is sufficient to prove the statement for any given model. Let V be the smooth locus of the affine k-variety defined by

$$(x_1^2 - ax_2^2)(y_1^2 - by_2^2)(z_1^2 - abz_2^2) = P(t).$$

Let Y be a smooth compactification of V with a projection $p : Y \to \mathbb{P}^1_k$. If one of the three numbers a, b, ab is a square in $k^\times$, this theorem is obvious since Y is rational. Then we only need to consider the case all numbers a, b, ab are not squares in $k^\times$.

We assume that Y has points in all completions of k, and we are given a finite set S of places of k containing all archimedean places, and points $P_v \in Y(k_v)$ for $v \in S$. We assume that there is no Brauer–Manin obstruction to weak approximation for $(P_v)_{v\in S}$.

By a similar argument as in the proof of Theorem 4.2, we can replace P_v by a point close to it in an open subset of Y for each finite place in S and 'move P_v to the infinite point' for each archimedean place v, then we obtain an open smooth affine subvariety U of Y defined by

$$(x_1^2 - ax_2^2)(y_1^2 - by_2^2)(z_1^2 - abz_2^2) = P^{(1)}(t) \neq 0,$$

such that P_v is contained in U for all $v \in S$ and P_v is very close to the smooth fiber of U at the infinite point for all archimedean places.

Let U_0 be the open subvariety of $\mathbb{A}^1$ defined by $P^{(1)} \neq 0$. Now we only need to look for a point $Q \in U_0(k)$ with the associated coordinate $\lambda \in k$, such that λ is very close to each $p(P_v)$ for any finite place $v \in S$, λ is big enough at each archimedean place of k (Q is close enough to the infinite point), and such that the fibre U_λ has a k-rational point.

Let $P^{(1)}(t) = cp_1(t)^{e_1} \cdots p_m(t)^{e_m}$, where $p_i(t)$ is irreducible over k. Let

$$A = \{(p_i(t), b) \in \mathrm{Br}(U) : 1 \leqslant i \leqslant m\} \cup \{(x_1^2 - ax_2^2, b)\}.$$

We know $(x_1^2 - ax_2^2, b)$ is the unique generator of the unramified Brauer group of the smooth fibre of Y (see [**7**, Theorem 4.1]). Obviously, $A \subset \mathrm{Br}(U)$.

We enlarge S so that it contains all the original places at which we want to approximate and that it also contains the places associated in Hypothesis H_1 (see [**18**, p. 71]) to the polynomials of $P^{(1)}(t)$ and all ramified places of K/k, where $K = k(\sqrt{a}, \sqrt{b})$.

There is no Brauer–Manin obstruction to weak approximation for $(P_v)_{v\in S}$. According to Harari's formal lemma (see [**21**]), we may find a finite set S_1 of places of k containing S, and points $P_v \in U(k_v)$, $v \in S_1$, and which extend the given family

$$P_v \in U(k_v), v \in S,$$

such that for each $\mathcal{B} \in A$

$$\sum_{v\in S_1} \mathrm{inv}_v(\mathcal{B}(P_v)) = 0. \tag{4.8}$$

Applying Hypothesis (H_1) (see [**18**, Proposition 4.1]), we thus find $\lambda \in k$ close enough to each $\lambda_v = p(P_v)$ for the finite places $v \in S_1$, λ integral away from S_1, and λ as large as we wanted at the archimedean places, such that:

(i) the fibre U_λ of p contains a k_v-point P'_v which is as close as we wish to P_v for all places $v \in S_1$, and such that

$$\mathrm{inv}_v(\mathcal{B}(P'_v)) = \mathrm{inv}_v(\mathcal{B}(P_v))$$

for each $\mathcal{B} \in A$ and $v \in S_1$;

(ii) for each irreducible term $p_i(t)$ of $P^{(1)}(t)$, there exists a place v_i such that $p_i(\lambda)$ is a uniformizer at v_i and is a unit at v if $v \notin S_1$ and $v \neq v_i$.

If $v \notin S_1$, then K/k is unramified at v. Then one of a, b, ab is a square in $k_v^\times$. Then the fibre U_λ contains a k_v-point P'_v for all places $v \notin S_1$.

Let $\mathcal{B} = (x_1^2 - ax_2^2, b)$. In the following, we will show $\mathrm{inv}_v(\mathcal{B}(P'_v)) = 0$ for any $v \notin S_1$:

(a) Suppose $v \neq v_i$ for $1 \leqslant i \leqslant m$.
  (1) If $k(\sqrt{b})/k$ is split at v, then $\mathrm{inv}_v(\mathcal{B}(P'_v)) = 0$.
  (2) If $k_v(\sqrt{b}) = k_v(\sqrt{a})$ is inert over k_v, then $\mathrm{inv}_v(\mathcal{B}(P'_v)) = 0$ obviously.
  (3) If $k_v(\sqrt{b})/k_v$ is inert and $k_v(\sqrt{a})/k_v$ is split, then

$$(y_1^2 - by_2^2, b)_v = (z_1^2 - abz_2^2, b)_v = 0.$$

One has

$$\mathrm{inv}_v(\mathcal{B}(P'_v)) = (x_1^2 - ax_2^2, b)_v = (P^{(1)}(\lambda), b)_v = 0$$

since $P^{(1)}(\lambda)$ is a unit at v.

(b) Suppose v is some v_i. For each $(p_i(t), b) \in A$, by (4.8) and point (i), we have

$$\sum_{v \in S_1} \mathrm{inv}_v(p_i(\lambda), b) = 0.$$

By global class field theory, we have

$$\sum_{v \notin S_1} \mathrm{inv}_v(p_i(\lambda), b) = 0. \tag{4.9}$$

By point (ii), we have $\mathrm{inv}_v(p_i(\lambda), b) = 0$ for $v \notin S_1$ and $v \neq v_i$, since $p_i(\lambda)$ is a unit at v. Then (4.9) implies $\mathrm{inv}_{v_i}(p_i(\lambda), b) = 0$. Since $p_i(\lambda)$ is a uniformizer at v_i, we have $k(\sqrt{b})/k$ is totally split at v_i, hence $\mathrm{inv}_{v_i}(\mathcal{B}(P'_{v_i})) = 0$.

By above arguments, we have

$$\sum_{v \in \Omega_k} \mathrm{inv}_v(\mathcal{B}(P'_v)) = \sum_{v \in S_1} \mathrm{inv}_v(\mathcal{B}(P'_v)) = 0$$

by (4.8) and point (i) for the last equality. Note that $\mathcal{B}$ generates the unramified Brauer group $\mathrm{Br}_{ur}(U_\lambda)/\mathrm{Br}_0(U_\lambda)$, there is no Brauer–Manin obstruction on U_λ for $(P'_v)_{v \in \Omega_k}$. There is a k-point on U_λ which is close enough to $(P'_v)_{v \in S}$ by [**35**, Theorem 8.12]. □

In the above, we considered the case that the extension K/k is abelian. In the following, we will consider the case that K/k is non-abelian. The original idea owes to Colliot-Thélène and the further development owes to Wittenberg.

THEOREM 4.6. *Let k be a number field and K/k non-Galois extension of prime degree p. Let K^{cl} be the Galois closure of K/k and F/k the maximal abelian subextension of K^{cl}/k. Let $P(t)$ be a non-zero polynomial over k. For any irreducible factor $p(t)$ of $P(t)$, let $L = k[t]/(p(t))$. We suppose*

$$F \not\subset L, \text{ or } F \subset L \text{ and } K.L/L \text{ is abelian}; \tag{4.10}$$

for example, if $p = 3$, any non-zero polynomial $P(t)$ satisfies (4.10). Assume Schinzel's hypothesis holds. Then the Brauer–Manin obstruction to the Hasse principle and weak approximation for rational points is the only obstruction for any smooth proper model of the variety over k defined by the equation

$$N_{K/k}(\Xi) = P(t).$$

Before the proof of Theorem 4.6, we need a lemma, which is similar to [**22**, Lemma 4.8].

LEMMA 4.7. *If v is a place of k which is unramified in K^{cl}/k and not totally split in F/k. Then the equation $N_{K/k}(\Xi) = a$ is solvable over k_v for any $a \in k_v^\times$.*

Proof. Write $K \otimes_k k_v := K_{w_1} \oplus \cdots \oplus K_{w_s}$. First, we will show $s > 1$.

If $s = 1$, then $K \otimes_k k_v = K_w$ is a field extension of k_v of degree p. By the assumption, v is not totally split in F/k, therefore $F \otimes_k k_v = F_{u_1} \oplus \cdots \oplus F_{u_t}$ with $F_{u_1} \neq k_v$. Let $d := [F_{u_1} : k_v] > 1$. Since $\mathrm{Gal}(K^{cl}/k)$ is a subgroup of the symmetric group $\mathcal{S}_p$, we have $d < p$, hence K_w and F_{u_1}

are linearly disjoint over k_v. Therefore, the Frobenius at v in $\mathrm{Gal}(K^{cl}/k)$ is an element of order divisible by dp. However, $\mathcal{S}_p$ contains no such elements, so the case s = 1 is impossible.

Let $d_i := [K_{w_i} : k_v]$. Since $p = d_1 + \cdots + d_s$ is a prime number and $s > 1$, then $\gcd(d_1, \ddots, d_s) = 1$. Therefore, there exist integers $n_1, \ldots, n_s$ such that $1 = n_1 d_1 + \cdots + n_s d_s$. It follows that

$$a = \prod_{i=1}^{s} N_{K_{w_i}/k_v}(a^{n_i}) \in N_{K/k}(K \otimes_k k_v),$$

so we are done. □

Proof of Theorem 4.6. Let V be the smooth locus of the affine k-variety defined by $N_{K/k}(\Xi) = P(t)$. Let Y be a smooth compactification of V with a projection $p : Y \to \mathbb{P}^1_k$ defined by $(\Xi, t) \mapsto t$.

We assume that Y has points in all completions of k, and we are given a finite set S of places of k containing all archimedean places, and points $P_v \in Y(k_v)$ for $v \in S$. We assume that there is no Brauer–Manin obstruction to weak approximation for $(P_v)_{v\in S}$.

By a similar argument as in the proof of Theorem 4.2, we can replace P_v by a point close to it in an open subset of Y for each finite place in S and 'move P_v to the infinite point' for each archimedean place v, that is, we get an open smooth affine subvariety U of Y defined by

$$N_{K/k}(\Xi) = P^{(1)}(t) \neq 0,$$

such that P_v is contained in U for all $v \in S$ and P_v is very close to the smooth fiber of U at the infinite point for all archimedean places.

Let U_0 be the open subvariety of $\mathbb{A}^1$ defined by $P^{(1)}(t) \neq 0$. We are now looking for a point $Q \in U_0(k)$ with associated coordinate $\lambda \in k$, such that λ is very close to each Q_v for $v \in S$, v finite, λ is big enough at each archimedean place (Q is close enough to the infinite point), and such that the fibre U_λ has a k-rational point.

Let $P^{(1)}(t) = cp_1(t)^{e_1} \cdots p_m(t)^{e_m}$, where $p_i(t)$ is monic and irreducible over k. Let $L_i = k[t]/(p_i(t))$. By our assumption (4.10) for $P(t)$, it is easy to verify that

$$F \not\subset L_i, \text{ or } F \subset L_i \quad \text{and} \quad K.L_i/L_i \text{ is abelian.} \tag{4.11}$$

Define

$$T_1 = \{1 \leqslant i \leqslant m : F \not\subset L_i\} \quad \text{and} \quad T_2 = \{1 \leqslant i \leqslant m : F \subset L_i\}.$$

For $i \in T_1$, we choose a non-trivial character χ_i of $\mathrm{Gal}(F/k)$, which is non-trivial on the subgroup $\mathrm{Gal}(F/F \cap L_i)$. Denote

$$A_1 = \{(p_i(t), \chi_i) \in \mathrm{Br}(U) : i \in T_1\},$$
$$A_2 = \bigcup_{i\in T_2} \{\mathrm{Cor}_{L_i/k}(t - \eta_i, \psi) \in \mathrm{Br}(U) : \psi \in \mathrm{Hom}(\mathrm{Gal}(K.L_i/L_i), \mathbb{Q}/\mathbb{Z})\},$$

where η_i is a root of $p_i(t)$ in L_i.

Let S' be a finite set of places of k containing all the archimedean places and all the bad finite places in sight: finite places where one $p_i(t)$ is not integral, finite places where all $p_i(t)$ are integral, but the product $\prod_i p_i(t)$ does not remain separable when reduced modulo v, places ramified in K^{cl}/k.

We enlarge S such that $S \supset S'$ and that S contains all the original places at which we want to approximate, and that S also contains the places associated in Hypothesis H_1 (see [**18**, p. 71]) to the polynomials of $P^{(1)}(t)$ and all ramified places of K^{cl}/k. There is no Brauer–Manin obstruction to weak approximation for $(P_v)_{v\in S}$. According to Harari's formal lemma (see [**21**]), we may find a finite set S_1 of places of k containing S, and points $P_v \in U(k_v)$, $v \in S_1$, which

extend the given family

$$P_v \in U(k_v), v \in S,$$

such that for each $\mathcal{B} \in A_1 \cup A_2$

$$\sum_{v \in S_1} \mathrm{inv}_v(\mathcal{B}(P_v)) = 0.$$

We claim that for any $i \in T_1$, we can find a place $v_i \notin S_1$ of k and a point P_{v_i} in $U(k_{v_i})$ such that

$$\mathrm{inv}_{v_i}(p_i(\lambda_{v_i}), \chi_i) \neq 0 \quad \text{and} \quad \mathrm{inv}_{v_i}(\mathcal{B}(P_{v_i})) = 0$$

for any $\mathcal{B} \in A_1 \cup A_2$ and $\mathcal{B} \neq (p_i(t), \chi_i)$, where λ_{v_i} is the coordinate of the image of P_{v_i} in $\mathbb{A}^1$. Therefore, we extend S_1 to $S_1 \cup \{v_i : i \in T_1\}$, then we have

$$\sum_{v \in S_1} \mathrm{inv}_v(\mathcal{B}(P_v)) \neq 0 \quad \text{for } \mathcal{B} \in A_1 \tag{4.12}$$

and

$$\sum_{v \in S_1} \mathrm{inv}_v(\mathcal{B}(P_v)) = 0 \quad \text{for } \mathcal{B} \in A_2. \tag{4.13}$$

We now prove this claim. For $i \in T_1$, let $E_i = F.L_i^{cl}$, where L_i^{cl} is the Galois closure of L_i/k. Let $\mathrm{Res}_{k/L_i}(\chi_i)$ be the restriction of χ_i to $\mathrm{Gal}(F.L_i/L_i)$, which is a non-trivial character by the choice of χ_i. Obviously, there exists $g \in \mathrm{Gal}(E_i/L_i)$ with $\mathrm{Res}_{k/L_i}(\chi_i)(\bar{g}) \neq 0$, where $\bar{g}$ is the image of g in the quotient group $\mathrm{Gal}(F.L_i/L_i)$. There are infinitely many places v of k such that g is contained in the conjugation class of the Frobenius of v by Chebotarev's density theorem. We can choose such a place v_i with $v_i \notin S$. Therefore, $p_i(t) = \pi_{v_i}$ has a solution $\lambda_{v_i} \in \mathfrak{o}_{k_{v_i}}$ and $p_j(\lambda_{v_i}) \in \mathfrak{o}_{k_{v_i}}^{\times}$ for $j \neq i$, where π_{v_i} is a uniformizer of k_{v_i}. Since v_i is not totally split in E/k, by Lemma 4.7, $U_{\lambda_{v_i}}$ has a k_{v_i}-point P_{v_i}. Therefore, we have

$$\mathrm{inv}_{v_i}(p_i(\lambda_{v_i}), \chi_i) \neq 0 \quad \text{and} \quad \mathrm{inv}_{v_i}(\mathcal{B}(P_{v_i})) = 0$$

for any $\mathcal{B} \in A_1$ and $\mathcal{B} \neq (p_i(t), \chi_i)$. Suppose $\mathcal{B} \in A_2$, since $p_j(\lambda_{v_i})$ is a unit, we also have $\mathrm{inv}_{v_i}(\mathcal{B}(P_{v_i})) = 0$.

Now apply Hypothesis (H_1) (see [**18**, Proposition 4.1]), we thus find $\lambda \in k$ close enough to each $\lambda_v = p(P_v)$ for the finite places $v \in S_1$, λ integral away from S_1, and λ as large as we wanted at all archimedean places, such that:

(i) the fibre U_λ of p contains a k_v-point P'_v which is as close as we wish to P_v for all places $v \in S_1$, and such that

$$\mathrm{inv}_v(\mathcal{B}(P'_v)) = \mathrm{inv}_v(\mathcal{B}(P_v))$$

for each $\mathcal{B} \in A_1 \cup A_2$ and $v \in S_1$;

(ii) for each irreducible term $p_i(t)$ of the polynomial $P^{(1)}(t)$, there exists a place v'_i such that $p_i(\lambda)$ is a uniformizer at v'_i and is a unit at v if $v \notin S_1$ and $v \neq v'_i$.

In the following, we will show that U_λ also contains a k_v-point for $v \notin S_1$. Since $\mathrm{Br}(U_\lambda) = \mathrm{Br}_0(U_\lambda)$ (see [**14**, Proposition 9.1]), there is a k-point on U_λ which is close enough to $(P'_v)_{v \in S}$ by [**35**, Theorem 8.12].

Suppose $v \notin S_1$ and $v \neq v'_i$ for $1 \leqslant i \leqslant m$. Since $P^{(1)}(\lambda)$ is a unit in k_v, U_λ contains a k_v-point.

Let $i \in T_1$. For each $\mathcal{B} \in A_1$, by (4.12) and point (i), we have

$$0 \neq \sum_{v \in S_1} \mathrm{inv}_v(\mathcal{B}(P_v)) = \sum_{v \in S_1} \mathrm{inv}_v(\mathcal{B}(P'_v)).$$

Then we have

$$\sum_{v\in S_1} \mathrm{inv}_v(p_i(\lambda), \chi_i) \neq 0.$$

By global class field theory, we have

$$\sum_{v\notin S_1} \mathrm{inv}_v(p_i(\lambda), \chi_i) \neq 0. \tag{4.14}$$

By point (ii), we have $\mathrm{inv}_v(p_i(\lambda), \chi_i) = 0$ for $v \notin S_1$ and $v \neq v_i'$, since $p_i(\lambda)$ is a unit at v. Then (4.14) implies

$$\mathrm{inv}_{v_i'}(p_i(\lambda), \chi_i) \neq 0.$$

Since $p_i(\lambda)$ is a uniformizer at v_i', we have F/k is not totally split at v_i'. By Lemma 4.7, U_λ has a $k_{v_i'}$-point for $i \in T_1$.

Let $i \in T_2$. For any $\mathcal{B} = \mathrm{Cor}_{L_i/k}(t - \eta_i, \psi) \in A_2$, by equation (4.13) and class field theory, similarly we have

$$0 = \sum_{v\notin S_1} \mathrm{inv}_v(\mathrm{Cor}_{L_i/k}(\lambda - \eta_i, \psi)) = \mathrm{inv}_{v_i'}(\mathrm{Cor}_{L_i/k}(\lambda - \eta_i, \psi)),$$

which is equivalent to

$$\sum_{w\in\Omega_{L_i}, w|v_i'} \mathrm{inv}_w(\lambda - \eta_i, \psi) = 0. \tag{4.15}$$

Since $\mathrm{ord}_{v_i'}(p_i(\lambda)) = 1$, in the decomposition of $L_i \otimes_k k_{v_i'}$ into a product of local fields $k_{i,w}$, $\lambda - \eta_i$ goes to a unit into all local fields $k_{i,w}$ but one, call it k_{i,w_i}, which is of degree 1 over $k_{v_i'}$ in which $\lambda - \eta_i$ becomes to a uniformizer. In (4.15), all terms but one then vanish. Thus the remain one $\mathrm{inv}_{w_i}(\lambda - \eta_i, \psi)$ also vanishes. Since $\lambda - \eta_i$ is a uniformizer in L_{i,w_i} and ψ runs through all characters of $\mathrm{Gal}(K.L_i/L_i))$, $K.L_i/L_i$ is totally split at w_i. Therefore, K can embed into $L_{i,w_i} = k_{v_i'}$, hence U_λ has a $k_{v_i'}$-point for $i \in T_2$. □

Remark 4.8. If $k = \mathbb{Q}$ and $P(t)$ is a product of (possibly repeated) linear factors over $\mathbb{Q}$, without assuming Schinzel's hypothesis, then similar results to Theorems 4.5 and 4.6 have been proved recently (see [**2**, **22**]).

5. *Brauer–Manin properties for zero-cycles of degree 1*

This section is devoted to the proof, for zero-cycles of degree 1, of unconditional versions of the theorems in Section 4, using Salberger's device, as in [**17**]. The same general comments as made in the beginning of the previous section may be made here, some related works were also performed in [**25**, **26**, **41**]. In particular, most of the results in the present section are not covered by [**17**, Theorem 4.1].

Theorem 5.1. *Let k be a number field and K/k an abelian extension. Let $P(t)$ be a non-zero polynomial over k. Let V be the smooth locus of the affine k-variety defined by*

$$N_{K/k}(\Xi) = P(t).$$

Let $T = R^1_{K/k}(\mathbb{G}_m)$. Suppose $Ш^2_\omega(\hat{T})_P = Ш^2_\omega(\hat{T})$ (see Section 1 for definition). If there is no Brauer–Manin obstruction to the existence of a zero-cycle of degree 1 on a smooth proper model V^c of V, then there is a zero-cycle of degree 1 on V^c (defined over k).

Proof. Let U be the smooth affine variety over k defined by

$$N_{K/k}(\Xi) = P(t) \neq 0.$$

Let U^c be a smooth compactification of U with a projection $p : U^c \to \mathbb{P}^1_k$.

Let X be the *CTHS* partial compactification (see Section 1 for definition) of V. By the assumption, we know $Ш^2_\omega(\hat{T})_P = Ш^2_\omega(\hat{T})$. Then we can choose a finite subset $B \subset \mathrm{Br}(X)$, such that the image of B by the composite map

$$\mathrm{Br}(X) \longrightarrow H^1(k, \mathrm{Pic}(\bar{X})) \longrightarrow Ш^2_\omega(\hat{T})$$

is $Ш^2_\omega(\hat{T})$ by the sequence (2.2).

Let $P(t) = cp_1(t)^{e_1} \cdots p_m(t)^{e_m}$, where $p_i(t)$ is a monic irreducible polynomial over k. Choose $\omega_1, \ldots, \omega_n \in \mathfrak{o}_K$ to be a basis of K over k, where $n = [K : k]$. Choose S to be a finite set of places of k containing all archimedean places such that $P(t) \in \mathfrak{o}_S[t]$ and $\omega_1, \ldots, \omega_n$ generate $\mathfrak{o}_K \otimes_{\mathfrak{o}_k} \mathfrak{o}_{k_v}$ over $\mathfrak{o}_{k_v}$ for any $v \notin S$. Let $\mathcal{U} \subset \mathbb{A}^{n+2}_{\mathfrak{o}_S}$ be the smooth affine integral model of U over $\mathfrak{o}_S$ defined by

$$N_{K/k}(x_1\omega_1 + \cdots + x_n\omega_n) = P(t) \quad \text{and } y \cdot P(t) = 1.$$

We enlarge S such that for any $\mathcal{B} \in B$, $\mathcal{B}$ can be extended to $\mathrm{Br}(\mathcal{U})$.

Let S_0 be a finite set of places of k containing S and all the bad finite places in sight: finite places where one $p_i(t)$ is not integral, finite places where all $p_i(t)$ are integral, but the product $\prod_i p_i(t)$ does not remain separable when reduced modulo v, places ramified in the extension K/k and all places v such that for some $\mathcal{B} \in B$, $\mathrm{inv}_w(j_K(\mathcal{B})) \neq 0$, where w is a place of K over v and $j_K : \mathrm{Br}(X) \to \mathrm{Br}(X_K)$, noting that $\mathrm{Br}(X_K) = \mathrm{Br}_0(X_K)$ by Lemma 4.1.

Let

$$A = \{(p_i(t), \chi) \in \mathrm{Br}(U) : 1 \leqslant i \leqslant m, \chi \in \mathrm{Hom}(\mathrm{Gal}(K/k), \mathbb{Q}/\mathbb{Z})\}.$$

Since we assume that there is no Brauer–Manin obstruction to the existence of a zero-cycle of degree 1, by an obvious variant of Harari's result [**18**, Theorem 3.2.2] we may find a finite set S_1 of places of k containing S_0 and for each $v \in S_1$ a zero-cycle z_v of degree 1 with support in $U \times_k k_v$ such that

$$\sum_{v \in S_1} \mathrm{inv}_v(\langle \mathcal{B}, z_v \rangle) = 0 \quad \text{for all } \mathcal{B} \in A \cup B.$$

Let s be the least common multiple of the orders of $\mathcal{B} \in A \cup B$. Let us write the zero-cycle z_v as $z_v^+ - z_v^-$, with z_v^+ and z_v^- effective cycles. Let z_v^1 be the effective cycle $z_v^+ + (ns-1)z_v^-$. We have $z_v = z_v^1 - nsz_v^-$, hence $\langle \mathcal{B}, z_v \rangle = \langle \mathcal{A}, z_v^1 \rangle$ since each $\mathcal{B}$ is killed by s. We thus have

$$\sum_{v \in S_1} \mathrm{inv}_v(\langle \mathcal{B}, z_v^1 \rangle) = 0 \quad \text{for all } \mathcal{B} \in A \cup B.$$

Let N_0 be a closed point of U such that $k(N_0) = K$. Similarly, $\langle \mathcal{B}, sN_0 \rangle = 0$. The degree of z_v^1 is congruent to 1 modulo ns. The cycle sN_0 has degree ns. Adding suitable multiples of sN_0 to each z_v^1 for v in the finite set S_1, then we find effective cycles z_v^2, all of the same degree $1 + Dns$ for some $D > 0$, and such that

$$\sum_{v \in S_1} \mathrm{inv}_v(\langle \mathcal{B}, z_v^2 \rangle) = 0 \quad \text{for all } \mathcal{B} \in A \cup B. \tag{5.1}$$

By the implicit function theorem and the continuity of the evaluation maps of the Brauer group, in (5.1), for each $v \in S_1$, each effective cycle z_v^2 may be assumed to be a sum of distinct closed points (that is, there are no multiplicities) whose images under $p_{k_v} : U_{k_v} \to \mathbb{A}^1_{k_v}$ are also distinct.

We claim that while keeping (5.1) we can moreover assume that, for each z_v^2 and each closed point P in the support of z_v^2, the field extension map $k_v(f(P)) \subset k_v(P)$ is an isomorphism.

Once more, it is enough to replace P by a suitable and close enough $k_v(P)$-rational point on $U_{k_v}(P)$: this follows from [**18**, Lemma 6.2.1].

Each of the zero-cycles $p(z_v^2)$ is now given by a separable monic polynomial $G_v[t] \in k_v[t]$ of degree $1 + Dns$, prime to $P(t)$ and with the property that the smooth fibres of p above the roots of G_v have rational points over their field of definition. By Krasner's lemma, any monic polynomial $H(t)$ close enough to $G_v(t)$ for the v-adic topology on the coefficients will be separable, with roots 'close' to those of G_v. Thus the fibres above the roots of the new polynomial are still smooth and still possess rational points over their field of definition.

An irreducible polynomial $G(t)$ of degree $1 + Dns$ defines a closed point M of degree $1 + Dns$ on $\mathbb{A}^1_k$. Let $F := k(M) = k[t]/(G(t))$. Let θ be the residue class of t in F. We now choose the irreducible polynomial $G(t)$ as given by [**17**, Theorem 3.1] with the field L in Theorem 3.1 contains K, and V' (V in Theorem 3.1) is the set of places of k at which L is totally split, such that

(i) θ is integral at $v \notin S \cup V'$;

(ii) for each place $v \in S_1$, $G(t)$ is close enough to $G_v(t)$, such that the fibre U_θ contains an F_w-point P_w for each place w of F over v, and such that $\sum_{w|v} P_w$ is 'close' enough to z_v^2 satisfying

$$\sum_{w|v} \mathrm{inv}_v(\mathrm{Cor}_{F_w/k_v}(\mathcal{B}(P_w))) = \mathrm{inv}_v(\langle \mathcal{B}, z_v^2 \rangle)$$

for each $\mathcal{B} \in A \cup B$;

(iii) for each irreducible term $p_i(t)$ of the polynomial $P(t)$, there exists a place w_i of F such that $p_i(\theta)$ is a uniformizer at w_i and is a unit at w if w is not over $S_1 \cup V'$ and $w \neq w_i$.

We claim that the fibre U_θ/F also has a point for w not over S_1.

(1) If the place w is not over $S_1 \cup V'$ and $w \neq w_i$ for all i, then it is clear that $\mathcal{U}_\theta$ has a $\mathfrak{o}_{F_w}$-point P_w since θ is integral at w by point (i) and $P(\theta)$ is a unit at w by point (iii).

(2) If the place w is over V', then it is clear that U_θ has a F_w-point P_w since $K.F/F$ is totally split at w.

(3) For each $\mathcal{B} = (p_i(t), \chi) \in A$, by (5.1) and point (ii), we have

$$\sum_{w|v, v \in S_1} \mathrm{inv}_v(\mathrm{Cor}_{F_w/k_v}(p_i(\theta), \chi)) = 0.$$

By global reciprocity law, one has

$$\sum_{w|v, v \notin S_1} \mathrm{inv}_v(\mathrm{Cor}_{F_w/k_v}(p_i(\theta), \chi)) = 0. \tag{5.2}$$

However, $\mathrm{inv}_w(p_i(\theta), \chi) = 0$ if w is not over S_1 and $w \neq w_i$, since either $p_i(\theta)$ is a unit in F_w or $K.F/F$ is totally split by point (iii). Hence (5.2) implies

$$\mathrm{inv}_{w_i}(p_i(\theta), \chi) = 0.$$

Since χ runs through all characters of $\mathrm{Gal}(K/k)$ and $p_i(\theta)$ is a uniformizer at w_i by point (iii), we have $K.F/F$ is totally split at w_i. Therefore, the fibre U_θ contains an F_{w_i}-point P_{w_i}.

We claim $\mathrm{inv}_w(\mathcal{B}(P_w)) = 0$ for any w is not over S_1.

(a) Suppose that w is a place of F over $v \notin S_1 \cup V'$ and $w \neq w_i$ for $1 \leqslant i \leqslant m$. For any $\mathcal{B} \in B$, we have $\mathrm{inv}_w(\mathcal{B}(P_w)) = 0$ since $\mathcal{B}$ can be extended to $\mathcal{U}$ and $P_w \in \mathcal{U}(\mathfrak{o}_{F_w})$.

(b) Suppose that w is some w_i or w is over V'. Then $K.F/F$ is totally split at w by the above argument (3) for $w = w_i$ and the definition of V' for w is over V'. Since $\mathrm{Br}(X_K) = \mathrm{Br}_0(X_K)$ by Lemma 4.1, the image of $\mathcal{B}$ in $\mathrm{Br}(X_{F_w})$ is trivial by the choice of S_0 ($S_1 \supset S_0$), hence $\mathrm{inv}_w(\mathcal{B}(P_w)) = 0$.

By (5.1), point (ii) and the above arguments, we have

$$\sum_{w\in\Omega_F} \mathrm{inv}_v(\mathrm{Cor}_{F_w/k_v}(\mathcal{B}(P_w))) = \sum_{v\in S_1} \mathrm{inv}_v(\langle \mathcal{B}, z_v^2\rangle) = 0.$$

Therefore,

$$\sum_{w\in\Omega_F} \mathrm{inv}_w(\mathcal{B}(P_w)) = 0$$

by the equality $\mathrm{inv}_v(\mathrm{Cor}_{F_w/k_v}\mathcal{B}(P_w)) = \mathrm{inv}_w(\mathcal{B}(P_w))$. Since $[F:k]$ and $[K:k]$ are relatively prime, we have $F\cap K = k$. Therefore, the natural map $\mathrm{Br}(T^c)/\mathrm{Br}_0(T^c) \to \mathrm{Br}(T^c_F)/\mathrm{Br}_0(T^c_F)$ is an isomorphism. By a similar argument as the last part in the proof of Theorem 4.2, B generates the unramified Brauer group $\mathrm{Br}_{ur}(U_\theta)/\mathrm{Br}_0(U_\theta)$. Therefore, there is no Brauer–Manin obstruction on U_θ for $(P_w)_{w\in\Omega_F}$, hence U_θ/F possesses an F-point by [**35**, Theorem 8.12]. □

With the help of Salberger's device, by similar argument as above we have the following result which corresponds to Theorem 4.5.

THEOREM 5.2. *Let k be a number field and $P(t)$ a non-zero polynomial over k. Let V be the smooth locus of the affine k-variety defined by*

$$(x_1^2 - ax_2^2)(y_1^2 - by_2^2)(z_1^2 - abz_2^2) = P(t),$$

where $a, b \in k^\times$. If there is no Brauer–Manin obstruction to the existence of a zero-cycle of degree 1 on a smooth proper model V^c of V, then there is a zero-cycle of degree 1 on V^c (defined over k).

Proof. Let U be the smooth affine variety over k defined by

$$(x_1^2 - ax_2^2)(y_1^2 - by_2^2)(z_1^2 - abz_2^2) = P(t) \neq 0.$$

Let U^c be a smooth compactification of U with a projection $p : U^c \to \mathbb{P}^1_k$. If one of the three numbers a, b, ab is a square in $k^\times$, then this theorem is obvious since Y is rational. Then we only need to consider the case all numbers a, b, ab are not squares in $k^\times$.

Write $P(t) = cp_1(t)^{e_1}\cdots p_m(t)^{e_m}$, where $p_i(t)$ is monic and irreducible over k. Let S_0 be a finite set of places of k containing all the archimedean places and all the bad finite places in sight: finite places where one $p_i(t)$ is not integral, finite places where all $p_i(t)$ are integral but the product $\prod_i p_i(t)$ does not remain separable when reduced modulo v, places ramified in the extension K/k, where $K = k(\sqrt{a}, \sqrt{b})$.

Let

$$A = \{(p_i(t), b) \in \mathrm{Br}(U) : 1 \leqslant i \leqslant m\} \cup \{(x_1^2 - ax_2^2, b)\}.$$

We know $(x_1^2 - ax_2^2, b)$ is the unique generator of the unramified Brauer group of the smooth fibre of U (see [**7**, Theorem 4.1]).

Since we assume that there is no Brauer–Manin obstruction to the existence of a zero-cycle of degree 1, by an obvious variant of Harari's result [**18**, Theorem 3.2.2] we may find a finite set S_1 of places of k containing S_0 and for each $v \in S_1$ a zero-cycle z_v of degree 1 with support in $U \times_k k_v$ such that

$$\sum_{v\in S_1} \mathrm{inv}_v(\langle \mathcal{B}, z_v\rangle) = 0 \quad \text{for all } \mathcal{B} \in A. \tag{5.3}$$

By the similar argument as in the proof of Theorem 5.1, for each $v \in S_1$, we may assume that:

(1) each cycle z_v is effective and has the same degree D, where D is odd;
(2) each cycle z_v is a sum of distinct closed points P_v whose images under $p_{k_v} : U_{k_v} \to \mathbb{A}^1_{k_v}$ are also distinct, and the field extension map $k_v(p(P_v)) \subset k_v(P_v)$ is an isomorphism.

Each of the zero-cycles $p(z_v)$ is now given by a separable monic polynomial $G_v[t] \in k_v[t]$ of degree D, prime to $P(t)$ and with the property that the smooth fibres of p above the roots of $G_v(t)$ have rational points over their field of definition.

An irreducible polynomial $G(t)$ of degree D defines a closed point M of degree D on $\mathbb{A}^1_k$. Let $F = k(M) = k[t]/(G(t))$. Let θ be the residue class of t in F. We now choose the irreducible polynomial $G(t)$ as given by [**17**, Theorem 3.1] with the field L in Theorem 3.1 contains K, and V' (V in Theorem 3.1) is the set of the places of k at which L is totally split, such that

(i) for each place $v \in S_1$, $G(t)$ is close enough to $G_v(t)$, such that the fibre U_θ contains an F_w-point P_w for each place w of F over v, and such that $\sum_{w|v} P_w$ is 'close' enough to z_v satisfying

$$\sum_{w|v} \mathrm{inv}_v(\mathrm{Cor}_{F_w/k_v}(\mathcal{B}(P_w))) = \mathrm{inv}_v(\langle \mathcal{B}, z_v \rangle)$$

for each $\mathcal{B} \in A$.

(ii) for each irreducible term $p_i(t)$ of the polynomial $P(t)$, there exists a place w_i of F such that $p_i(\theta)$ is a uniformizer at w_i and is a unit at w if w is not over $S_1 \cup V'$ and $w \neq w_i$.

If w is not over S_1, then $K.F/F$ is unramified at w. Then one of a, b, ab is a square in $F_w^\times$, hence the fibre U_θ contains an F_w-point P_w.

Let $\mathcal{B} = (x_1^2 - ax_2^2, b)$. In the following, we will show $\mathrm{inv}_w(\mathcal{B}(P_w)) = 0$ for any w is not over S_1.

(a) Suppose $w \neq w_i$ for $1 \leqslant i \leqslant m$.
  (1) If $F(\sqrt{b})/F$ is split at w, then $\mathrm{inv}_w(\mathcal{B}(P_w)) = 0$.
  (2) If $F_w(\sqrt{b}) = F_w(\sqrt{a})$ is inert over F_w, then $\mathrm{inv}_w(\mathcal{B}(P_w)) = 0$ obviously.
  (3) If $F_w(\sqrt{b})/F_w$ is inert and $F_w(\sqrt{a})/F_w$ is split, then

$$(y_1^2 - by_2^2, b)_w = (z_1^2 - abz_2^2, b)_w = 0.$$

  One has

$$\mathrm{inv}_w(\mathcal{B}(P_w)) = (x_1^2 - ax_2^2, b)_w = (P(\theta), b)_w = 0$$

  since either $P(\theta)$ is a unit at w or $K.F/F$ is totally split at w.

(b) Suppose w is some w_i. For each $(p_i(t), b) \in A$, by (5.3) and point (i), we have

$$\sum_{v \in S_1} \mathrm{inv}_v(\mathrm{Cor}_{F_w/k_v}(p_i(\theta), b)) = 0.$$

By global reciprocity law, one has

$$\sum_{v \notin S_1} \mathrm{inv}_v(\mathrm{Cor}_{F_w/k_v}(p_i(\theta), b)) = 0. \tag{5.4}$$

However, $\mathrm{inv}_w(p_i(\theta), b) = 0$ for w not over S_1 and $w \neq w_i$, since either $p_i(\theta)$ is a unit at w or $K.F/F$ is totally split at w by point (ii). Therefore, (5.4) implies

$$\mathrm{inv}_{w_i}(p_i(\theta), b) = 0.$$

Since $p_i(\theta)$ is a uniformizer at w_i, we have $F(\sqrt{b})/F$ is totally split at w_i, hence $\mathrm{inv}_{w_i}(\mathcal{B}(P_{w_i})) = 0$.

By the above arguments, we have

$$\sum_{w\in\Omega_F} \mathrm{inv}_w(\mathcal{B}(P_w)) = \sum_{w|v\in S_1} \mathrm{inv}_w(\mathcal{B}(P_w)) = \sum_{v\in S_1} \mathrm{inv}_v(\langle \mathcal{B}, z_v\rangle) = 0$$

by (5.3) for the last equality. Since $F\cap K = k$, we deduce that B generates the unramified Brauer group $\mathrm{Br}_{ur}(U_\theta)/\mathrm{Br}_0(U)$ by [**7**, Theorem 4.1]. Therefore, there is no Brauer–Manin obstruction on U_θ for $(P_w)_{w\in\Omega_F}$. Hence U_θ/F possesses an F-point by [**35**, Theorem 8.12]. □

For all primes p (do not need the condition (4.10)), we can prove the following result (which corresponds to Theorem 4.6).

THEOREM 5.3. *Let k be a number field and $P(t)$ a non-zero polynomial over k. Let p be a prime. Let K/k be of degree p and not cyclic. Let V be the smooth locus of the affine k-variety defined by*

$$N_{K/k}(\Xi) = P(t).$$

If there is no Brauer–Manin obstruction to the existence of a zero-cycle of degree 1 *on a smooth proper model V^c of V, then there is a zero-cycle of degree* 1 *on V^c (defined over k).*

Proof. Obviously, V has a closed point N_0 with $k(N_0) = K$. Then we only need to show that there is a field F/k with degree $[F : k]$ prime to p such that $V^c(F)\neq\emptyset$.

Let K^{cl} be the Galois closure of K/k with the Galois group G. Then G is a subgroup of the symmetric group $\mathcal{S}_p$. Let H be the p-Sylow subgroup of G. We can see H is cyclic and of order p. Let Θ be the fixed field by H. Then $[\Theta : k]$ is relative prime to p.

One has a family $\{z_v\}_v$ of zero-cycles of degree 1 which is orthogonal to $\mathrm{Br}(V^c)$. Then one pushes $\{z_v\}_v$ to V^c_Θ, one gets a family of zero-cycles of degree 1 on V^c_Θ. A projection formula for the Brauer pairing shows the family of zero-cycles on V^c_Θ is orthogonal to $\mathrm{Br}(V^c_\Theta)$. Since $K^{cl}/\Theta = K.\Theta/\Theta$ is cyclic (of degree p), one gets a zero-cycle of degree 1 on V^c_Θ by [**17**, Theorem 4.1]. So there is a field F/Θ with the degree $[F : \Theta]$ prime to p such that $V^c(F)\neq\emptyset$. Since $[\Theta : k]$ is prime to p, we have $[F : k] = [F : \Theta]\cdot[\Theta : k]$ is also prime to p. □

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